Geometry of Confabulation and Refusal in Small Language Models

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https://github.com/vkazei/confabqa

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Abstract

Faced with an unanswerable question, small language models (SLMs) can either refuse or confabulate: confidently produce a wrong answer. Do SLMs really know when confabulating? Can we flip confabulations into refusals? The hidden state just before the first generated token is the well-known primary window into an LM brain. I map the geometry of this space with linear probes, a sparse autoencoder (SAE), and direct optimization of intervention directions, in depth on Qwen3-1.7B and cross-examined on Gemma 2 2B and Llama 3.2 3B, where an analogous late-layer refusal direction appears. On Qwen I closely inspect three candidate directions for the refusal vector: probe-recovered, SAE feature, and directly optimized. A push at or below the state's own norm along these directions flips most confabulations into refusal openings dominated by the "As…" family. The opposite push collapses refusals mostly into confabulations, yet leaves originally correct answers intact, proving the usefulness of steering along found directions. Confabulations do not resolve into a concentrated feature in a dictionary or a token signature on Qwen3-1.7B.

The dataset and evaluation design break two typical structural issues in confabulation studies. ConfabQA is a 784-item synthetic benchmark whose obscure pre-cutoff category breaks the well-known correctness/knowledge-cutoff confound. Scoring every probe on the margin over the strongest of four prompt-feature baselines extracts the hidden state's own contribution. Correctness signal is model-dependent: only for Llama 3.2 3B it is portable: 81% of the probe's margin over test majority survives transfer between [Confab, Pop, Trivia]QA. I release ConfabQA, the judge, the analysis code, and the cached activations. Steering in hidden space flips a certain number of confabulations to refusal openings, yet only 30% are marked as true refusals by the judge. Further research is needed to make this a robust mechanism.

1. Introduction

When a small language model (SLM) is asked a factual question about something it never learned, it either refuses or answers anyway. A fraction of those answers happen to be correct; the rest are confabulations: fluent, confident, and wrong. The term, argued for over "hallucination" by Smith et al. (2023), names pattern-based reconstruction the model does not recognize as wrong. Can we read that choice off the hidden state, and steer it? Kadavath et al. (2022) argued that LLMs "mostly know what they know"; a probing literature followed (Azaria and Mitchell 2023; Marks and Tegmark 2024), fitting linear classifiers on hidden states and reporting correctness-prediction accuracies in the 80–90% range, with behavioral detectors (sampling consistency, Manakul et al. 2023; semantic entropy, Farquhar et al. 2024) as the black-box complement. A high probe accuracy is not by itself evidence that the hidden state knows anything. In adjacent settings, recent work traces above-chance accuracy to the prompt rather than the model's internal state: introspection (Singh et al. 2026) and reasoning mode (Sahoo et al. 2026). A confabulation probe faces the same question. Does it read the model's state, or the question?

I map the geometry of this decision with linear probes, a sparse autoencoder, and direct optimization of intervention directions, and test whether steering along the recovered directions turns confabulations into refusals, under controls that isolate the hidden state's own contribution. I introduce **ConfabQA**, a probing benchmark of 4 domains $\times$ 3 categories (well-known pre-cutoff, *obscure* pre-cutoff, post-cutoff); the third category, items whose answers were in the training data but which the model is expected to fail, populates the cell standard pre-vs-post-cutoff designs leave empty, breaking the cutoff/correctness confound. ConfabQA, plus 800-item samples of PopQA (Mallen et al. 2023) and TriviaQA (Joshi et al. 2017), run through three instruction-tuned models in the 1.7–3B range (Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B) with Qwen3-4B as a within-family scaling control; every probe is scored against both the majority baseline and a stricter question-text baseline, the control that decides whether the hidden state contributed anything.

Relation to prior work. The prompt-feature baseline used here is a *selectivity control* in the lineage of Hewitt and Liang (2019), who showed that probe accuracy without a control task mostly measures the probe rather than the representation (Belinkov 2022 surveys the failure modes). Such controls remain rare in the correctness-probing literature, where time-sensitive question sets routinely conflate "post-cutoff" with "model doesn't know." The closest mechanistic analogue is probably Ferrando et al. (2025), whose *entity-recognition latents* in Gemma-2 SAEs causally steer between answers and refusals; the present work recovers the same class of feature via a probe + SAE pipeline on a different model family with a disconfounded benchmark (§6.3), and adds a cross-dataset transfer test of content-invariance (§8.3).

Findings. Two complementary results emerge:

1. *Refusal vs. confabulation is a model-internal feature: each model linearly encodes its own upcoming speech act.* On the wrong+refusal subset of ConfabQA the late-layer probe beats the strongest prompt baseline on every model: $+7.4\%$ on Qwen3-1.7B, $+16.3$ on Gemma 2 2B, $+2.2$ on Llama 3.2 3B (the Llama margin is within fold-to-fold spread; its powered counterpart is the Section 8.2 refusal-channel gap). That some such signal exists is natural (the label is the model's own output decision); the findings are its *form*: a single linear direction in the late layers (Gemma peaks mid-network, at 19 of 26), causally sufficient for the first-token refusal opening, and decomposable. A one-shot hidden-state intervention drives the first-token refusal-opener rate to 100% on Qwen3 and Gemma (Sections 6.2 and 7.2), though only about a third of these openings complete as judged refusals; the rest revert to content, and closing that gap is left to future work. This replicates Arditi et al.'s (2024) single-direction finding in the epistemic-abstention rather than safety-refusal regime (mean-difference steering lineage: Panickssery et al. 2024; answerability readable from pre-generation states: Slobodkin et al. 2023; known-unknowns benchmarks: Yin et al. 2023), with the cutoff variable controlled by construction: the contribution is the cutoff control, the causal completion, and the decomposition, a sparse-autoencoder resolution of the direction into a deferential-register ensemble (canonical refusal-opener plus apology, officialese, and hedging registers; Section 6.3) of which the opener feature alone flips $30/30$ wrong-item next-token argmaxes to refusal openers. The reverse search finds no counterpart: no concentrated dictionary feature or token signature picks out confabulation, which behaves as the default attempt pathway missing a fact (Section 6.3).
2. *Llama 3.2 3B carries a substantially larger and more general correctness signal than Qwen3-1.7B or Gemma 2 2B.* On balanced PopQA / TriviaQA subsamples, Llama's probe beats the prompt baseline by $+21$–$+25\%$, $5.7\times$ and $2.2\times$ Qwen's margins on the same benchmarks. Three controls localize the source: not parameter count (the Qwen3-4B scaling control moves the margin by less than 2%; Section 8.1); not dataset-specific overfit (Llama keeps $\approx 81\%$ of its margin under cross-dataset transfer without refit, where Qwen's and Gemma's transfer collapses; Section 8.3); not refusal-channel readout ($\approx 83\%$ survives dropping refusals, and an independent refusal probe adds a further $+18$–$+25\%$ gap; Section 8.2). Gemma is behaviorally intermediate, yet its correctness probe is the most dataset-local of the three; the most plausible single explanation is Llama's heavy calibrated-abstention training forcing a content-invariant "model is uncertain" representation that neither Qwen3's nor Gemma's training developed.

2. Theoretical background

The subject models are decoder-only transformers with causal attention: the final-layer state at the last position is projected through the language-model (LM) head onto the vocabulary $\mathcal{V}$, and generation feeds each emitted token back as input (Figure 8a, Appendix B.6); the *prefill* pass (Figure 8b), the single forward pass over the prompt alone before any token is generated, is the pass the probes read.

Qwen3-1.7B instantiates this with $L = 28$ blocks, hidden size $d = 2048$, and $|\mathcal{V}| \approx 152{,}000$ tokens with tied input/output embeddings (checkpoint and settings in §4).

2.1 Confidence metrics: log-probability and entropy

Two scalar confidence summaries of the generated response recur in §5, both means over the generated positions: $\bar{\ell}$, the mean log-probability of the emitted (argmax) tokens, so that $\exp(\bar{\ell})$ is the geometric mean of the per-token probabilities ($\bar{\ell} = -0.05$ corresponds to ≈ 0.95 per token), and $\bar{H}$, the mean Shannon entropy of the next-token distribution; all generation is greedy (`do_sample=False`, deterministic), so the emitted token is the mode. Formal definitions (Eq. (2)), the worst-step diagnostic $\min_t \ell_t$, and the empirical anti-correlation of the two are in Appendix B.6.

2.2 The prefill hidden state as the canonical going-in representation

The *prefill* pass over the prompt $x_{1:T}$ produces the full hidden-state tensor; I pick off the **hidden state at every layer at the last prompt position**, $\{h_T^{(\ell)}\}_{\ell=0}^{L}$, of shape $(L+1, d) = (29, 2048)$ per question. Nothing has been generated yet, so this is the model's representation *immediately before it commits to the answer* (Figure 8b): it summarizes the whole prompt, sits one linear projection from the first answer token's logits (Section 4), and avoids the contamination of probing generated tokens' states (expanded rationale in Appendix B.6).

2.3 Why a linear probe (and the specific pipeline I use)

Following Alain and Bengio (2017), I fit at each layer ℓ a binary linear classifier on $\{h_T^{(\ell)}\}$; accuracy substantially above the appropriate baseline implies a hyperplane separates the property at depth ℓ, while whether the model *uses* that direction is a separate question. The pipeline is `StandardScaler -> PCA(n_components=16) -> LogisticRegression(max_iter=2000, C=1.0)` under 5-fold stratified CV with `random_state=0`, reported as mean and per-fold std of accuracy; scaler and PCA are fit only on the training fold, so no test-fold leakage occurs (overfit rationale and schematic in Appendix B.6/B.7, Figure 7). Appendix D.1 sweeps $n_{\text{components}}$ over $\{4, 8, 16, 32, 48, 64\}$ to show the headline numbers are not specific to 16.

2.4 The two confounds and what counts as evidence

Confound 1 (cutoff/correctness). A benchmark of easy pre-cutoff plus post-cutoff questions makes "correct" and "pre-cutoff" nearly the same label, so a correctness classifier is in effect a cutoff classifier ("this question mentions 2025"). ConfabQA disconfounds with *obscure pre-cutoff* questions (facts in the training data the model still gets wrong), so pre-cutoff no longer implies correct (keeping only pre-cutoff questions would also remove the refusals; Section 5.1); the disconfound test fits the correctness probe restricted to pre-cutoff items, where accuracy above the pre-cutoff majority cannot be explained by the cutoff variable.

Confound 2 (hidden state/prompt features). A probe that beats its majority baseline might still be reading question-text features any classifier could extract: year, domain, length, keywords. The honest baseline is a logistic regression on prompt-side features alone (roster in Section 4, specification in Appendix B.7, schematic in Figure 9, Appendix B.6); a hidden-state probe is evidence of model-internal information only to the extent it beats this baseline outside per-fold noise (h_{adds} = probe peak − strongest baseline, in pp, reported throughout §§5–8). This is the control the probing literature most often omits: it distinguishes "the hidden state encodes self-knowledge" from "the hidden state encodes the prompt, which encodes the answer."

3. Sourcing and curation of the ConfabQA question set

ConfabQA comprises 784 items in a 4×3 design: 4 domains (science, history, culture, cinema) crossed with 3 categories (well-known pre-cutoff, obscure pre-cutoff, post-cutoff), totalling 143 well-known, 153 obscure, and 488 post-cutoff items (per-cell counts in Table 3, Appendix A). Items are generated from per-domain templated source files; each carries a provenance URL for its gold answer and a validation status field populated by an external LLM with web access, and the third of three validation iterations, **ConfabQA** ($n = 784$), is what the paper analyzes (full design in `data/QUESTIONS_v1_CARD.md`; pipeline reproducibility and a sample source entry in Appendix B.2 and B.5). A fifth domain (sports) produced $0/26$ correct, leaving no positive examples for the within-pre-cutoff probe; it is excluded and reported in Appendix A.1.

The third (`obscure`) category is the design innovation: it populates the cell "answer-was-in-training-data but model-is-expected-to-fail," which the standard two-cell design leaves structurally empty. A typical obscure item asks who won the Best Actor Academy Award at the 1968 ceremony (gold: Rod Steiger, on the public record); Qwen3-1.7B confidently answers with the wrong actor.

4. Methods

Subject model and generation. Qwen3-1.7B (`Qwen/Qwen3-1.7B`; Qwen Team 2025), a 1.72B-parameter text-only instruction-tuned decoder-only transformer (Apache 2.0), 28 blocks, hidden dimension 2048, run in bfloat16 on Apple Silicon MPS in 16 GB unified memory; greedy decoding throughout (`do_sample=False`, `max_new_tokens=64`, `enable_thinking=False`, suppressing Qwen3's `<think>` trace), so all generations are deterministic conditional on the prompt.

Hidden-state capture. Each question runs `model.generate(...)` with `output_hidden_states=True` and `output_scores=True`, persisting one $(L+1, d) = (29, 2048)$ prefill tensor per question. One indexing fact matters downstream: entries $1..L-1$ of the returned stack are raw post-block residuals, but the final entry (index $L = 28$) is the *final normed* state (HF applies the last RMSNorm before recording it), so the probes read the normed state at "layer 28" while the Section 6.2/6.3.1 hooks and the SAE operate on the pre-norm post-block-27 residual.

Same-model judge. Each generated answer is labeled by Qwen3-1.7B itself, called separately with a structured prompt (`judge.py`) asking for one of `correct`, `refusal`, or `wrong` (decision rule, worked examples, and parser in Appendix B.1); the judge replaces the brittle substring-match grader used in earlier work.

Judge agreement is measured on the full benchmark by three LLM annotators applying the same three-way rule strictly (Table 9, Appendix B.3): the Qwen3-1.7B judge, Claude Opus 4.7 on a stratified 30-item sample, and Google Deep Research (DR; Gemini 3.1 Pro with web-search verification) covering all 784. DR vs. the Qwen-judge gives $\kappa = 0.78$, "substantial agreement" (Landis and Koch 1977); Claude agrees $30/30$, bounding the Qwen-judge's disagreement with a second independent grader at $\leq 10\%$ at 95% confidence by the rule of three; DR is systematically stricter, reflecting its containment rule and gold verification rather than judge miscalibration (decomposition and retained flagged-premise items in Appendix B.3). A sensitivity rerun under the DR labels keeps the picture, and a blind Claude-family regrade of the Section 6 intervened generations agrees with the self-judge without inflating flip rates (both in Appendix B.3).

Hidden-state probe pipeline. For each binary target and layer, the Section 2.3 pipeline is fit on the last-prompt-token states at that depth, reported as mean and std of fold accuracy plus a "layer range", the set of layers within 1σ of the peak.

Prompt-feature baseline pipeline. For each target, I also fit baselines with no access to the hidden state, reporting 5-fold CV accuracy on the *same folds* (`random_state=0`). Four baselines of increasing leniency: **TF-IDF** (bag-of-tokens logistic regression on the raw question text); **text-only** (engineered numeric features); **text + domain** (adds the domain one-hot); and **text + domain + category** (adds the *annotator's* expected-difficulty dummy, the least conservative baseline; for the `cutoff` target it is excluded from the strongest-baseline comparison, since the dummy would be the label itself). Full feature lists and hyperparameters are in Appendix B.7. The honest comparison is the hidden-state probe vs. the *strongest* of the four on each target: the "h adds" column of Table 7 in Section 5.6.

Probe targets. `correct` (all 784 items); `cutoff` (post vs. pre, all 784); `refusal_vs_wrong` (the `{refusal, wrong}` subset, n=549); `refusal_vs_wrong_within_post` (n=393; all 147 refusals are post-cutoff, so this conditions out the cutoff covariate); `correct_within_pre` (n=296, the cutoff disconfound test); and `correct_within_obscure` (n=153, the popularity disconfound test).

Bootstrap protocol (Sections 8.1, 8.2). For each `(dataset, model, target)` cell, draw $K = 30$ random 50/50 class-balanced subsamples without replacement; each refits the full probe pipeline (peak over all 29 layers) plus the four prompt baselines on the same folds, and $\bar{h}_{\text{adds}}$ is reported with a percentile 95% CI (class-size rule, CI construction, and compute bounds in Appendix E). One asymmetry when reading small margins: the probe side maximizes over 29 correlated layers, the baseline side over four, so under a no-signal null the expected h_{adds} is slightly positive; margins within a couple of points of zero carry this selection bias.

External-dataset evaluation (Section 8.1). **PopQA** (Mallen et al. 2023) and **TriviaQA** (Joshi et al. 2017) run end-to-end through the same generation pipeline and three-way Qwen3-1.7B judge,

$n = 800$ per seed (splits, strata, seeds in Appendix E); three seeds per `(subject model, dataset)` widen the pool to $n_{\text{unique}} \approx 2{,}200$, Llama runs are single-seed ($n_{\text{unique}} = 800$), and the judge remains Qwen3-1.7B in all runs (Section 10 records the Qwen-judge-on-Llama-outputs caveat).

5. Probing Qwen3-1.7B

Before the numbers, the Confabulation Atlas (Figure 5, Appendix A.4) situates them geometrically: a *supervised* 2-D projection of the layer-18 hidden states of all 784 questions, against an *unsupervised* PCA(2) control in which PC1+PC2 alone already support an 86.9% refusal-vs-wrong probe (vs. 73.2% majority), so the refusal separation is not an artifact of supervised projection.

5.1 Accuracy by stratum

Of 784 items, 235 (30.0%) are graded `correct`; 147 (18.8%) `refusal`; 402 (51.3%) `wrong` (breakdowns in Appendix A, Table 2). The cutoff manipulation works as designed: pre-cutoff accuracy is 47.3% (140/296) against 19.5% (95/488) post-cutoff, and in the obscure-pre-cutoff cell, the benchmark's innovation, the model fails on $\approx 66.7\%$ of items whose answers existed in its training data, exactly the population the cutoff disconfound needs. All 147 refusals fall in the post-cutoff cell: refusal is triggered by the knowledge-cutoff cue rather than intrinsic uncertainty, motivating the `refusal_vs_wrong_within_post` probe of Section 4.

5.2–5.4 Text-level confidence, confident-confabulation case studies, and hidden-state geometry

Mean per-token log-probability separates correct from incorrect answers only in expectation (Figure 4, Appendix A.2): $\bar{\ell}_{\text{correct}} \approx -0.10$ vs. $\bar{\ell}_{\text{incorrect}} \approx -0.15$ (≈ 0.90 vs. ≈ 0.86 geometric-mean per-token probability), with heavy overlap in $[-0.2, -0.05]$: weakly informative, not a usable predictor. Table 6 (Appendix A.3) lists the five highest-$\bar{\ell}$ items per failure label; the five confabulations all sit at ≈ 0.95 geometric-mean per-token probability, and refusals are paradoxically *high-confidence* text outputs (fluent, assertive statements of ignorance), so a binary substring grader lumps both failure modes into one `incorrect` bucket and text-level confidence cannot separate them afterward: separating them is what the three-way judge is for (gold-side validation case study in Appendix B.2). Unsupervised PCA(2) of the raw hidden state (Figure 6, Appendix A.4) shows the classes mixing heavily at layer 14 while by layer 28 the refusals pull into a distinct cluster with no supervision, a tight group of confident confabulations separating out on its own: that late-layer refusal cluster is the geometry the refusal-vs-wrong probe reads.

5.5 Per-layer linear probes against majority baselines

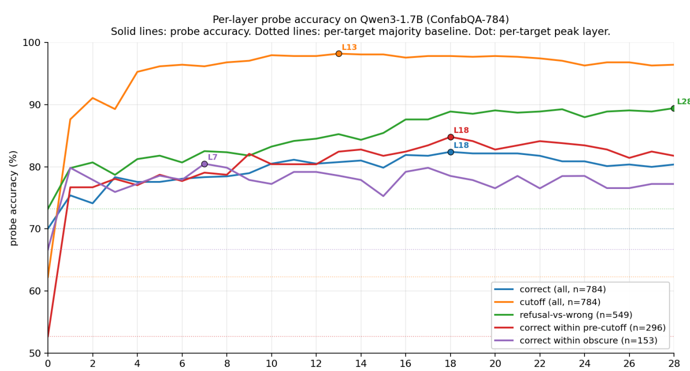

Figure 1: Per-layer probe accuracy for all five targets on Qwen3-1.7B (ConfabQA). Solid lines: probe accuracy per layer; dotted horizontal lines: per-target majority baselines; dots: peak layers (values in Table 8, Appendix A.5). Cutoff saturates early and stays high; refusal-vs-wrong rises monotonically and peaks at the deepest block; correctness and correct-within-pre peak mid-network, with within-pre's headline margin over its pre-cutoff majority largely vanishing against the prompt-feature baseline of Section 5.6; correct-within-obscure has a 1-σ layer range spanning nearly the whole network.

Table 8 (Appendix A.5) tabulates peaks, layer ranges, and margins over the majority baselines for all five targets, with the 1σ layer-range caveat (the within-obscure range spans the whole network, so its per-layer peak is not a per-depth claim). Taken at face value these are the margins a single-model study would lead with; Section 5.6 explains why every correctness margin among them is overstated.

Imbalance-aware metrics at the refusal-vs-wrong peak (balanced accuracy 0.874, ROC AUC 0.944; confusion matrix in Appendix D.6) show the probe is not defaulting to "wrong".

5.6 Value of the hidden state compared to prompt features

Table 7 scores the hidden-state probe against the four Section 4 baselines on the same folds. The `+category` baseline is a *difficulty oracle*: category explains most of the raw correctness variance by construction (the $62.2\%/33.3\%/19.5\%$ base rates of Table 2), so scoring against it asks for item-level signal beyond the coarse difficulty bucket; within single-category strata the dummy is constant (`+category` collapses to `+domain`), and for `cutoff` the dummy *is* the label and is excluded (Section 4).

Full per-target margins over the strongest prompt baseline are in Table 7 (Appendix A.5).

(All baselines are 5-fold CV accuracy with `random_state=0`. †: for the `cutoff` target the category dummy *is* the label; the tautological 1.000 is shown for completeness but excluded from the strongest-baseline comparison.)

Table 7 interpretation. (1) *Correctness gains little from the hidden state*: $\approx 3\%$ or less over the strongest baseline ($+2.4$ all items, $+3.0$ within-pre, -2.5 within-obscure), within per-fold noise (0.031–0.063); the Section 8.1 bootstrap confirms the reading (all-items margin real, disconfounded within-stratum margins consistent with zero). (2) *Cutoff is easy to predict either way*: text features alone reach 94.0–95.2% (the year mentioned is a near-perfect predictor), the probe at 98.2% adds $+3.0\%$; a manipulation check, not a finding about internals. (3) *Refusal-vs-confabulation substantially beats prompt features*: $+7.4\%$ over TF-IDF overall, $+9.6\%$ within post-cutoff where all refusals occur, gaps ≈ 4x the per-fold std (0.019, 0.022). The asymmetry is what the targets' nature predicts: refusal-vs-confabulation is a decision the model itself makes and the late hidden state is the computation producing it (Section 2.2), so an encoding must exist; correctness is a relation between output and world, and nothing guarantees the model represents it. What is not guaranteed, and is the substance of Section 6, is the signal's form: linearly decodable from 16 principal components, concentrated late, causally sufficient for the first-token opening, and SAE-decomposable.

6. From probe to tokens: the Qwen3-1.7B refusal direction

The probes of Section 5 certify that a refusal signal exists beyond prompt features; this section asks what the signal *is*: *read* the direction through the model's own output head (§6.1), *push* it onto real hidden states to test causation (§6.2), and *take it apart* into sparse dictionary features to test which part carries the causal effect (§6.3).

6.1 Probe-direction atlas: what the refusal signal points at in token space

The refusal-vs-wrong probe at layer 28 carries the largest baseline-adjusted margin of Table 7. Its weight vector lifts back to raw hidden-state units, $(\mathbf{w}_{\text{raw}})_i = (V^\top w)_i/\sigma_i$ (PCA basis $V \in \mathbb{R}^{16\times 2048}$, scaler stds σ, sign chosen so the refusal end is positive), and is pushed through the model's final RMSNorm and tied LM head: $\ell_v = \left[\text{LMHead}(\text{RMSNorm}(\mathbf{w}_{\text{raw}}))\right]_v$ for $v \in \mathcal{V}$. RMSNorm is scale-invariant, so the ℓ_v depend only on the direction: the same linear readout the head applies to any real state, so moving a state along the direction adds approximately these scores to its next-token logits, a prediction Section 6.2 confirms, with the lens scores as the $\alpha \to \infty$ asymptote of the dose-response (calibration in Appendix C). One caution: this is *epistemic* abstention expressed as a hedging opener, a different object from the *safety*-refusal direction of Arditi et al. (2024); the two need not coincide, and nothing here tests whether they do.

Top tokens (full subset, $n = 549$). The ten highest-scoring tokens are all tokenizations of "as", and the first non-"as" token, at rank 11, is `there`, the second-most-common refusal opening (scores in Appendix D.4); the refusals near-universally open with "As of my knowledge cutoff in 2023…", so the probe is reading exactly that output-token commitment. An opener is a propensity, not a guarantee, which is why the causal test scores the strict judge label alongside the first-token rate. The confabulation pole shows no thematic structure, and the layer-18 *correctness* direction (all $n = 784$ items) shows no comparable lens structure either (token detail in Appendix D.4); Section 6.2 tests its causal counterpart.

The n=393 within-post recovery (where every refusal actually lives) yields the same vocabulary in sharper form (Appendix D.4); with the within-post probe's $+9.6\%$ margin (Section 5.6), the late-layer state carries a discrete pragmatic signal ("I am about to hedge") not extractable from the question text (score histograms in Figure 11, Appendix D.4). A recovery-method check (Appendix D.4, Figure 2) shows the raw class-mean difference used by safety-refusal work is *not* this direction: it absorbs topical and temporal-currency content where the probe concentrates the register-and-opener component. In sum: a direction whose token-level signature is the refusal-opening vocabulary, a late-network commitment to a specific speech act, not a general "self-knowledge" representation.

6.2 Causal validation by direct intervention

The probe direction of Section 6.1 is *correlational*. To test whether it is also *causal*, a forward hook adds $\alpha \cdot \mathbf{w}_{\text{unit}}$ to the last prompt token's post-block-27 residual during prefill on a stratified subset (30 wrong post-cutoff + 30 refusal items), where $\mathbf{w}_{\text{unit}}$ is the unit-normalized within-post recovery of Section 6.1 (hook detail in Appendix C); generation passes through unmodified, one-shot at the moment of commitment. The perturbed residual is *not* the cached normed state the probes read (Section 4); a coordinate-dissociation check refitting the probe natively on pre-norm residuals finds equal discriminative accuracy but half the causal leverage (Appendix C), so the normed representation, one linear map from the output head, is where the potent direction lives. RMSNorm sets the natural α scale; working doses run from a third of the pushed state's own norm to about twice it (dose ratios and calibration scan in Appendix C). **Sweep:** $\alpha \in \{-2000, -500, 0, 500, 1500, 3000\}$ on the 60 items, greedy, re-judged three-way; outcomes are the strict judge-label refusal rate and a first-token refusal-opener rate over the Section 6.1 opener set (listed in Appendix C). The opener rate is deliberately a first-token *proxy* (an "As…" opening does not commit the model to refusing), which is why both measures are reported and diverge.

Table 1: The steering matrix: outcome rates for one-shot pushes along the refusal direction, doses as fractions of the mean pre-norm state norm ($\|\bar{\mathbf{h}}\| \approx 1640$; sweep $\alpha \in \{-2000, -500, 0, +500, +1500, +3000\}$); $n = 30$ per subset. Judged CORRECT stays 0%–10% on the wrong subset and judged REFUSAL $\leq 14\%$ on the correct subset throughout, for every direction; the small judged-REFUSAL rates at negative doses on wrong items ($\leq 13\%$, 0% opener rate) are sign-inconsistent magnitude artifacts, not directional effects.

original label	metric	$-1.2\times$	$-0.3\times$	0	$+0.3\times$	$+0.9\times$	$+1.8\times$
WRONG	opener rate	0%	3%	10%	50%	97%	100%
	judged REFUSAL	10%	0%	0%	13%	30%	30%
REFUSAL	opener rate	0%	23%	100%	100%	100%	100%
	judged REFUSAL	17%	40%	100%	100%	100%	100%
	judged WRONG	77%	53%	0%	0%	0%	0%
CORRECT	opener rate	0%	0%	0%	10%	63%	100%
	judged CORRECT	93%	100%	100%	87%	57%	53%
	judged WRONG	7%	0%	0%	7%	30%	33%

Table 1 is the steering matrix. Pushed toward refusal, wrong items open with refusal vocabulary almost without exception from a $0.9\times$ dose, yet only a third of full generations convert: the autoregressive continuation pulls back to content. Pulled away, refusing items collapse into attempts that are overwhelmingly confabulations; correct items are untouched by the pull and largely resist the push, about half still delivering the fact through a forced refusal opening. Suppressing the refusal decision does not unlock knowledge, and pushing toward refusal does not reliably erase it: the direction gates the opening speech act, and whether the fact is present decides what follows. The matrix is a property of the refusal axis, not of one recovery: rerun with the SAE-2191 and directly optimized directions, it reproduces the same plateaus row for row (parity block, Table 10, Appendix C).

Interpretation. The direction is a causal handle on the refusal-opening *readout*: it localizes where the commitment is written, not the upstream circuitry computing it (transported earlier-layer pushes are future work). The full outcome is *co-determined* by autoregressive dynamics the one-shot push does not touch (a model can open "As of 2024…" and still commit to a fact, hence the 30% vs. 100% gap on wrong items); the reverse direction is stronger because the natural refusal trajectory is itself committed. A prefix-forcing control (Appendix C) reproduces the judge-label outcome without touching the hidden state: the direction causally selects the first token, and everything after token one is ordinary autoregressive dynamics (sample generations in Appendix C). This is the epistemic-abstention counterpart of *shallow safety alignment* (Qi et al. 2025), where the safety-alignment delta concentrates in the first few output tokens; here the abstention decision is likewise written, and steerable, at token one.

Pushing the correctness direction the same way finds no comparable causal handle: effects are magnitude-driven and sign-symmetric, the few flips concentrate on a recurring handful of marginal items, and nothing indicates the direction injecting knowledge; this correctness-direction null (protocol, rates, flip forensics, and the tension with truthfulness-steering reports) is detailed in Appendix C. The asymmetry with Table 1 is what the two behaviors demand: refusing is a speech act the model can always produce, so a single direction can carry it; answering correctly requires the specific fact, which a direction can at most dislodge from a marginal default, not supply.

6.3 SAE feature decomposition of the refusal direction

The logit lens says what the direction outputs; a sparse autoencoder (SAE), trained on a far larger general-text corpus run through the model (so its features reflect what matters *to the model*, not what varies in one dataset), helps answer how the decision is composed. For a hidden state $h \in \mathbb{R}^{2048}$, the TopK SAE (Gao et al. 2025) computes

$$a(h) = \mathrm{Enc}(h) \in \mathbb{R}^{32{,}768}_{\geq 0}, \quad \|a(h)\|_0 = L_0 = 50, \quad h \approx \mathrm{Dec}(a(h)) = \sum_{f:\, a_f(h)>0} a_f(h)\, \mathbf{d}_f.$$

Each state is represented by 50 of $32{,}768$ candidate features; each feature f owns a fixed decoder direction $\mathbf{d}_f$ in the same space as h and $\mathbf{w}_{\text{raw}}$, and remains a descriptive coordinate unless it passes a causal test, as 2191 does in Section 6.3.1. The SAE is Qwen-Scope for Qwen3-1.7B-Base (`qwen-scope-3-1.7b-base-w32k-l50`; Qwen Team 2026; the Qwen analogue of Gemma Scope; Lieberum et al. 2024), applied at its declared hook, the post-block-27 residual (Section 4); it was trained on the *base* model while the subject is *Instruct* (reconstruction checks in Appendix D.2; PCA-coverage and blend tests in Appendix D.3). Three *correlational* views rank which features compose the direction: direct encoding, decoder alignment, and the empirical refusal-vs-wrong activation differential ($n = 147$ vs. $n = 402$; definitions in Appendix D.7); feature 2191 is the only one in the top eight of all three views, and the apology feature 14034 appears in all three top-twenty lists.

In the SAE's own geometry the refusal direction decomposes not into one opener plus content detectors but into a coherent *deferential-register ensemble* (feature card in Figure 15, Appendix D.7, with per-feature hit rates and vocabularies): a `moment` hedge (4314, the strongest discriminator), deferential officialese (17077), polite modality (14361), the canonical opener itself (2191), a second `Please` feature (16612), a first-person feature (8875), and the apology register (14034), in that order of refusal-vs-wrong selectivity. **Feature 2191**'s decoder lens is *exactly* the §6.1 refusal vocabulary (`as`, 作为, 作為, `\tas`, `As`, `as`, `As`); it fires on 100% of refusals (mean activation 268) and 49% of wrongs, and it is the ensemble's causal anchor (Section 6.3.1). **Feature 14034** (`Sorry`/`Oops` register) fires on 97% of refusals, yet its apology vocabulary never surfaces in the outputs, and pushing its decoder direction at doses that saturate 2191 produces zero `Sorry`/`Oops` openers (Appendix C); 17077 behaves the same way: register markers, not emitted tokens. The "refusal direction" is thus a superposition of register features jointly encoding *how the model is about to speak*, of which exactly one (2191) writes the opener token that carries the causal effect; the register is represented far more richly than it is spoken. The reading beyond 2191 is correlational (only the opener member is causally tested), and under a dictionary reconstructing these states at EV 0.50 feature identities carry splitting and absorption risk (Chanin et al. 2025); how an h-space push maps into a-space is analyzed in Appendix D.4.

The wrong pole has no commitment feature. The mirror search over all $32{,}768$ features finds nothing like 2191: the strongest wrong-side features are *topic*, phrasing, and attempt-formatting cues, not commitment markers (ranked features and hit rates in Appendix D.7); the dictionary contains a refusal register but no confabulation marker. The claim is scoped to what was searched (one dictionary at one hook, per-item EV 0.50, so absence of a dictionary feature is not absence of signal; a probe on the dictionary's residual $h - \mathrm{Dec}(a(h))$ still separates the classes at probe-grade accuracy, Appendix D.2). Confabulation behaves as the default attempt pathway missing a fact, not a marked state, the a-space form of the Section 5.6 prompt-feature story.

6.3.1 Causal validation of feature 2191

The ensemble decomposition is *correlational*. Rerunning the Section 6.2 one-shot protocol with the feature's decoder direction substituted for the probe direction (on 30 originally-wrong items, reading off the next-token probability of the opener set, 10 unique token IDs), the dose-response is monotonic with a sharp transition between $\alpha = 200$ and $\alpha = 750$: from $P = 0$ to $P = 1$ across two $\log_2$ steps, and from $0/30$ to $30/30$ argmax tokens flipped to refusal openers (Clopper–Pearson 95% CI $[0.88, 1.00]$; $n = 30$ is small, so true rates as low as ≈ 0.88 are consistent). Both sweeps add unit-norm vectors, so doses are directly comparable: the feature saturates at $\alpha = 750$ ($0.46\times$ the mean state norm) where the probe direction needs roughly twice the dose (three-direction dose-response table in Appendix C, Table 12); a single SAE feature is the more dose-efficient causal handle, causally *sufficient* for refusal-opener generation on wrong items (the refusal-item baseline at $\alpha = 0$ is already $P = 0.969$).

Two scope notes: 2191 was selected by the correlational views before any intervention ran, so the $30/30$ evaluation is not circular (though the wrong items come from the same ConfabQA pool as the selection statistics). Three norm-matched random directions plus a shuffled-label probe direction leave the steering matrix unmoved at sub-norm doses (rates in Appendix C; one random seed partially collapses refusals at a super-norm dose, so super-norm doses can act by magnitude alone, which is why the working regime stays at or below the state norm).

Two consequences. The §6.1 lens story is mechanically grounded (the SAE recovers exactly the opener vocabulary the lens projected to), and the probe direction's causal effect runs through 2191 specifically; its ensemble neighbors (14034, 17077, 4314) contribute to the *correlational* direction only. The two vectors are far from the same object, cosine 0.16 to the full-subset recovery and 0.25 to the within-post direction actually pushed (chance ≈ 0.02 in 2048 dimensions), yet produce the same first-token flip, 81° apart in the plane they span (Figure 12, Appendix D.4; norm accounting in Appendix C).

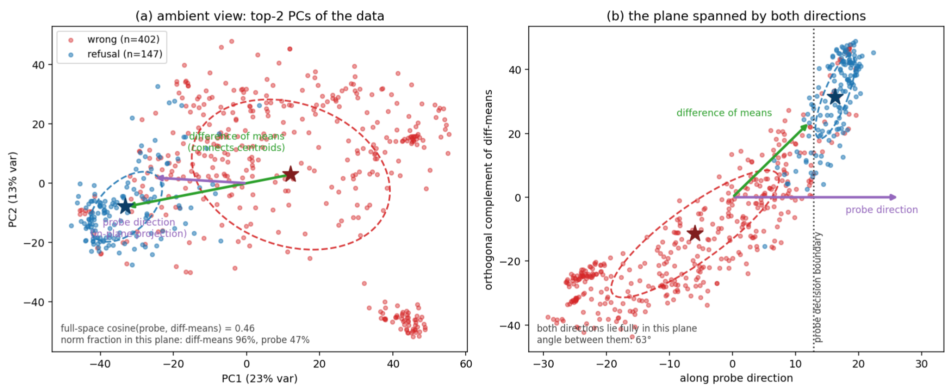

Figure 2: Two recoveries of "the" refusal direction (layer-28 refusal+wrong states, $n = 549$; stars: class centroids; dashed curves: per-class 1σ covariance ellipses). **(a)** Ambient view, top two principal components: the class-mean difference (green) connects the centroids and lies almost entirely in this top-variance plane; the probe direction (purple, in-plane projection) does not. **(b)** The plane spanned by the two directions, in which both lie fully: the angle between them is 63°. The wrong class's covariance ellipse is elongated along the same oblique axis the centroid connector rides, while the probe direction is the axis whose marginal separates the classes item by item (score histograms in Figure 11). The dotted vertical line is the probe's decision boundary, exact in this panel because the hyperplane normal is the probe direction itself.

A third recovery: the optimized refusal direction. Where the probe and the class-mean difference recover the direction from *data*, a third route recovers it from the *objective*, maximizing the opener log-probability under the final RMSNorm and LM head with the vector as the only trainable parameter (setup, cosines, and lens in Appendix C). The optimum lands next to feature 2191, not the probe direction, and is the most dose-efficient of the three (Table 12); the ordering optimized, then SAE feature, then probe matches how directly each recovery is shaped by the output objective, and what the probe direction adds on top of the opener cone is discriminative signal, not causal leverage. Figure 3 shows the whole geometry at once.

All four causal routes converge on the strict criterion: on the same wrong items, the probe push, the 2191 push, and prefix-forcing either literal opener produce the same downstream judged-refusal rate of roughly one in three, consistent with complete mediation at the first token (per-route rates, the not-powered-as-equivalence caveat, and dose-for-dose accounting in Appendix C).

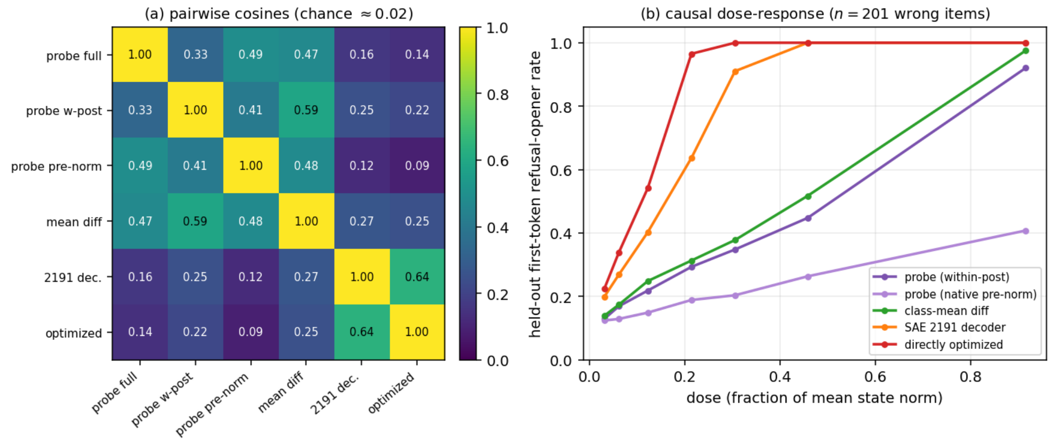

Figure 3: The geometry of the six refusal-direction candidates. **(a)** Pairwise cosines split the recoveries into two families: data-side (probe variants and the class-mean difference, 0.33–0.59) and output-side (2191 decoder and directly optimized, 0.64), weakly coupled (0.09–0.27; chance ≈ 0.02 in 2048-d). **(b)** Held-out first-token flip curves, doses as fractions of the mean state norm: causal potency tracks output-family membership, not discriminative quality. The native pre-norm probe separates the classes as well as any recovery (Appendix C) yet carries the least leverage.

7. Cross-model results

Do the Qwen3-1.7B findings survive a change of model? This section reruns the full pipeline on **Gemma 2 2B** (`unsloth/gemma-2-2b-it`) and **Llama 3.2 3B** (`unsloth/Llama-3.2-3B-Instruct`), then the causal intervention with each model's own recovered refusal direction.

7.1 Comparison (Qwen3-1.7B, Gemma 2 2B, Llama 3.2 3B)

All outputs are graded by the same judge (Qwen3-1.7B) for a standardized correctness threshold; Table 16 (Appendix D.5) compiles the key statistics, alongside the per-layer probe curves for all three models (Figure 14).

The three models reach nearly identical overall accuracy (30–32%) but adopt sharply different abstention policies on post-cutoff questions: Llama refuses almost every post-cutoff failure (97.5%, $475/487$), Qwen only about a third, Gemma in between; this behavioral axis, not accuracy, is what the refusal probes track. The correctness margins collapse to a few percentage points against the "+category" baseline on all three models (Section 5.6's confound, cross-model; depth-band detail in Appendix D.5), while refusal is the axis on which the three most cleanly separate: the refusal-vs-wrong probe beats the strongest prompt baseline by $+7.4\%$ (Qwen), $+16.3\%$ (Gemma), and $+2.2\%$ (Llama), peaking in the deepest layers on Qwen and Llama (layer 28) and at layer 19 on Gemma, the late-layer localization the Section 7.2 intervention exploits.

7.2 Causal intervention

The Section 6.2 protocol applied to Gemma 2 2B and Llama 3.2 3B, using each model's own recovered refusal direction at its own probe peak layer (Gemma: 19; Llama: 28); judge = Qwen3-1.7B for all three runs (Gemma's chat template lacks a system role and cannot self-judge), and α ranges are calibrated per model to its hidden-state scale (Table 13, Appendix C). The order-of-magnitude spread in operational α is itself a finding: the refusal direction is causal in each model, but its operational range is set by the model's hidden-state scale and tolerance to off-manifold perturbation.

Gemma 2 2B, whose refusal probe carried the largest margin over its prompt baseline, is also the most manipulable: the wrong-to-refusal flip reaches $0\% \to 87\%$ (vs. Qwen3's $0\% \to 30\%$) at the sweet-spot $\alpha = +2000$ with 100% opener rate (sweep and off-manifold pathologies in Table 14, Appendix C).

Llama 3.2 3B's near-saturated default refusal policy (97.5% of post-cutoff failures) leaves only $n = 12$ confabulating items, insufficient for a clean directional-causal claim. Its tightly normalized hidden state goes off-manifold symmetrically at $|\alpha| \geq 300$. The cleanest directional signal is the reverse: the refusal rate drops from 100% to 80% at $\alpha = -300$ ($3/15$ previously-refusing items committing to a confabulation), a -20% shift that is real but small, consistent with Llama's $+2.2\%$ refusal-probe margin (full sweep, pathologies, and examples in Appendix C). Because the recorded Gemma sweep ran at 6–25$\times$ its residual scale, a sub-norm rerun with controls was added.

At $0.31\times$/$0.92\times$ of Gemma's state norm the real direction converts 30%/40% of wrong items to judged refusals, monotonically in dose, versus 3–20% for two norm-matched random directions (cleanest at $0.31\times$: 30% vs. 3–7%). Gemma's refusals themselves resist sub-norm de-refusal (26–$30/30$ hold; artifact `17_intervention_gemma_subnorm`). The intervention is thus causal in Qwen and Gemma at doses at or below their state norms, with random-direction controls on both, and directionally suggestive on Llama; a model's default abstention policy bounds both the *signal magnitude* of its refusal-vs-wrong probe and the *operational dynamic range* of the intervention.

8. Cross-dataset results

Do they survive a change of question distribution? This section moves from ConfabQA to PopQA and TriviaQA: bootstrap confidence intervals with a within-family scaling control, a refusal-channel attribution, and a no-refit transfer test.

8.1 Balanced-subsample bootstrap (ConfabQA, PopQA, TriviaQA)

Are the single-seed h_{adds} point estimates robust to sampling noise, and do the conclusions transfer to broader factual benchmarks? The Section 4 bootstrap protocol runs on **nine** ConfabQA cells plus **five** external-dataset cells (PopQA $\times$ {Qwen3-1.7B, Qwen3-4B, Llama 3.2 3B} + TriviaQA $\times$ {Qwen3-1.7B, Llama 3.2 3B}), fourteen `(dataset, model, target)` combinations (label provenance in Appendix E); the Qwen3-4B PopQA cell is the within-family scaling control isolating parameter count from family / post-training recipe.

Reading the fourteen cells (Table 17 and forest plot, Appendix E). 5 of the 9 ConfabQA cells, and $5/5$ external-dataset cells, have 95% CIs excluding 0; the Qwen3 and Gemma disconfounded within-stratum cells remain consistent with zero but with positive point estimates in all four cases, so the single-seed "null" reading was borderline rather than emphatic. The CIs are per-cell and unadjusted for multiple comparisons; the cross-model contrast rests on no single borderline cell. Two protocol checks calibrate them (Appendix E): a label-permutation null shows $\approx 1\%$-range margins should not be read as evidence while the all-items margins sit far above the null, and an item-level bootstrap leaves both Llama within-stratum cells consistent with zero, so those two positive CIs should be read as point estimates.

Cross-dataset $\times$ cross-model contrast. On the same PopQA data under the same substring-correct criterion (Qwen pooled over three seeds, Llama single-seed; Section 4), Llama recovers a $+24.9\%$ correctness margin where Qwen3-1.7B recovers $+4.4\%$, a $5.7\times$ gap on $1.8\times$ the parameter count; on TriviaQA the gap is $2.2\times$, and it is driven neither by raw accuracy nor by the prompt-baseline floor (accounting in Appendix E): Llama's hidden state is qualitatively richer in linearly-decodable correctness information.

Within-family scaling control (Qwen3-4B). Qwen3-4B (same family, $2.4\times$ Qwen3-1.7B's parameter count, $\approx 25\%$ more than Llama 3.2 3B) lands at $+5.77\%$, 95% CI $[+2.34, +11.61]$: statistically indistinguishable from Qwen3-1.7B's $+4.35\%$ and far below Llama's $+24.94\%$. Doubling parameter count within the Qwen family moves $\bar{h}_{adds}$ by $\approx 1.4\%$; switching family at smaller parameter count moves it by $\approx 19\%$: **the cross-model gap is not a parameter-count effect**; it tracks model family / post-training recipe (corroborating raw accuracies in Appendix E; artifact list in Appendix B.4).

8.2 Refusal-channel attribution

The up-to-$5.7\times$ h_{adds} gap invites a mechanical explanation: the probe target labels "wrong" together with "refusal," and Llama refuses far more than Qwen on these benchmarks (PopQA 62.9% vs. 1.7%; census in Appendix E), so its h_{adds} could be refusal-channel readout rather than factual self-knowledge: the "refusal direction is the whole atlas" hypothesis, extended to external data.

Two tests run under the same bootstrap protocol on the four cells with appreciable refusal counts: *Test A* (drop refusals) filters the pool to `judge_label` $\in$ {`correct`, `wrong`}, balances 50/50, and re-probes, so any surviving h_{adds} is genuine correct-vs-wrong discrimination among attempted answers; *Test B* (probe refusal directly) targets `judge_label == 'refusal'` balanced 50/50, measuring how cleanly the abstention decision is encoded beyond prompt-text predictability.

Headline. Dropping refusals leaves Llama's margin largely intact ($+24.94 \to +20.76$ on PopQA, $+21.25 \to +17.71$ on TriviaQA): the refusal channel accounts for $\approx 17\%$ of the headline and the bulk ($\approx 83\%$) is real correct-vs-wrong discrimination on attempted items, while the Qwen rows are essentially unchanged (Qwen refuses too rarely here for the confound to bite). Probed directly, Llama's refusal target beats the strongest prompt baseline by $+18.44$ (PopQA) and $+25.43\%$ (TriviaQA) with CIs excluding 0 by wide margins; Qwen's analogous margins are positive but underpowered. One caveat: the ratio's numerator is judge-labeled and its denominator substring-labeled, and a label-sensitivity rerun shows the contrast is not a label artifact (full table, per-cell readings, and the rerun in Appendix E, Table 19). Llama 3.2 3B thus carries two independent linearly-decodable signals, correct-vs-wrong among attempted answers (Test A) and a clean abstention direction (Test B), with correctness the larger share: the "refusal direction is the whole atlas" framing was right about Qwen3 and wrong about Llama.

8.3 Transfer of the correctness probe

Is the probe direction itself portable: does a correctness probe fit on dataset A classify dataset B without refit, isolating a content-invariant signal from content-specific correlates? For each model the pipeline is fit on each dataset at the peak-within layer and evaluated on the other two *without refitting*: a 3×3 matrix, within-dataset CV on the diagonal, pure transfer off it, each cell scored against the test-dataset majority (layer grids, frozen-statistics detail, and the TF-IDF transfer control in Appendix E).

Headline. Averaging margins (probe-acc minus test-majority) across the 3 within cells and the 6 transfer cells (Table 18, Appendix E): Llama retains 81% of its within-dataset margin under transfer ($+18.9\%$ within, $+15.2\%$ transferred, $5/6$ transfers beating the test majority), while Gemma's retention is -54% (the largest within margin of the three, but two catastrophic collapses with PopQA as target) and Qwen's is -14% (an almost flat probe staying almost flat); per-model detail in Appendix E.

Reading. Within-dataset peak performance (Llama $>$ Gemma $>$ Qwen) and transfer portability (Llama $\gg$ Gemma $\approx$ Qwen) are *different orderings*: Gemma decodes dataset-specific content patterns while a Llama direction found on any pool classifies the others nearly as well as the source. The parsimonious explanation aligns with §1's finding 2: Llama's heavy calibrated-abstention training imposes a content-invariant "the model is uncertain" representation the other two trainings did not produce (full 3×3 matrices released; Appendix B.4).

9. Discussion

The picture is more model-dependent than the ConfabQA single-seed reading suggested. The disconfounded within-stratum margins are small on all three models (single digits against the strongest prompt baseline, one cell negative); the bootstrap leaves the Qwen and Gemma cells consistent with zero, and Llama's within-stratum cells are suggestive rather than established (Section 8.1). Llama's powered evidence is the external-dataset margins, whose CIs sit far from zero under either label regime, partitioned by Section 8.2 into mostly genuine correct-vs-wrong discrimination, a smaller refusal-channel share, and an independent clean abstention direction besides.

A tempting over-generalization, rejected. The Qwen3-1.7B results alone would support the framing "the hidden-state correctness probe adds nothing beyond a prompt-text classifier at 1.7B–3B scale"; on balanced external subsamples Llama's correctness probe adds $+17$–$+25\%$ over the strongest prompt baseline, 95% CIs excluding 0 by 14–17%. The stronger reading that the refusal direction is the *only* recoverable structure fails the same way: at 3B Llama scale, both refusal and correctness directions exist in legible form.

The robust positive result (consolidated in Appendix F): the refusal direction's logit lens recovers each model's literal refusal-opening vocabulary, the one-shot intervention saturates the first-token opener rate on Qwen3 and Gemma, the Section 8.2 refusal probe extends the finding to PopQA and TriviaQA at Llama scale, and the geometry is visible in unsupervised PCA, while the Qwen3-1.7B correctness direction fails all three corresponding tests. **Why two atlases on Llama and one on Qwen** (Appendix F): the most plausible mechanism is Llama's heavier calibrated-abstention training, with the Qwen3-4B control ruling out parameter count. **Implication** (expanded in Appendix F): probes reporting ~ 80–90% correctness prediction should be re-evaluated against a prompt-feature baseline, and the outcome is sharply model-dependent, so headline small-LM claims need at least two architectures with different post-training recipes and bootstrap CIs on balanced subsamples.

10. Limitations

1. **Benchmark construction.** Four known imperfections in ConfabQA, none of which affect the prompt-feature baseline (it conditions on text): a rater-subjective obscure/well-known boundary; gold answers as a validation snapshot (facts drift, the pipeline must be re-run); Anglophone source skew; and a fuzzy pre/post-cutoff boundary in the science domain, where Qwen3's post-training corpus apparently included a handful of post-cutoff OpenAI/Anthropic/SpaceX items, so the near-perfect cutoff probe partly recovers "what kind of entity" rather than a clean date discriminator.
2. **Evaluation methodology.** $n = 784$ gives only ≈ 157 items per CV fold, and the within-obscure probe's 1σ layer range spans the entire network; the prompt-feature baseline is hand-crafted and likely a *lower* bound on what a generic text classifier could extract; the bootstrap resamples balanced subsets, not the underlying pool, so its percentile CIs are subsample CIs, not population CIs; and the Section 8.3 transfer matrix covers the correctness probe only (Gemma's ≤ 5 and Qwen's few dozen external refusals, Table 19, are too few for a cross-dataset refusal fit, while Llama's counts would support one; a Llama refusal-transfer test plus a known-unknowns benchmark eliciting refusals from the smaller models would close the gap).
3. **Scope and judge generalization.** Four models in the 1.7B–4B range, no $\gtrsim 7$B replication; the scaling control rules out parameter count as the dominant driver of the cross-model h_{adds} gap, but family and post-training recipe remain confounded, and a clean isolation would swap the recipe holding family fixed. The Qwen judge on Qwen subject (and on Llama outputs in Section 8.2) is partially de-risked by the cross-model intervention runs (Gemma reproduces the Qwen refusal pattern under an external Qwen judge), but an independent or human re-grade on a Llama-output subset remains the cleanest remaining check.

11. Conclusion and future work

I introduce **ConfabQA** and apply it, plus PopQA and TriviaQA samples, to three instruction-tuned models and a Qwen3-4B within-family scaling control, with $K = 30$ balanced-subsample bootstrap CIs across 14 `(dataset, model, target)` cells. Findings:

(i) **The hidden state encodes a refusal-vs-confabulation signal that beats prompt features across all three models** (ConfabQA $+7.4$–$+16.3\%$; deepest layers; the logit lens recovers each model's literal refusal-opening vocabulary; causal under a one-shot forward-hook intervention on Qwen and Gemma; on Llama the sweep ran at $\approx 8\times$ its remeasured state norm and is directional evidence only).

(ii) **The correctness-probe gap is sharply model-dependent.** On balanced PopQA / TriviaQA subsamples Llama 3.2 3B adds $+24.9$ and $+21.3\%$ over the strongest prompt baseline (95% CIs $[20.6, 29.0]$, $[19.5, 22.9]$) where Qwen3-1.7B adds $+4.4$ and $+9.6\%$ (CIs $[2.1, 6.5]$, $[5.1, 13.8]$); Llama's ConfabQA disconfounded point estimates are also positive ($+11.4\%$ within-pre, $+12.3\%$ within-obscure) though an item-level bootstrap leaves both consistent with zero (Section 8.1). The "correctness probe reduces to prompt features at 1.7–3B scale" framing was an over-generalization from one model.

(iii) **Llama's correctness signal is primarily not refusal-channel readout:** dropping refusals leaves $+20.8\%$ (PopQA) and $+17.7\%$ (TriviaQA), so $\approx 83\%$ of the headline h_{adds} is genuine factual self-knowledge among attempted items, and a direct refusal probe additionally recovers $+18.4$–$+25.4\%$, an independent clean abstention direction. Llama at 3B carries **two** linearly-decodable signals; Qwen3-1.7B carries only weak versions of either at this scale and on these benchmarks.

The methodological contribution is the joint use of a prompt-feature baseline, $50/50$ class-balanced subsampling, percentile-CI bootstraps over $K = 30$ such subsamples, and cross-model plus cross-dataset evaluation; a single-model, single-seed, class-imbalanced read of the same code could have reported either polarity. Directions for v2: (i) swap the post-training recipe holding family fixed and check whether the cross-family h_{adds} gap survives at $\gtrsim 7$B; (ii) extend the one-shot intervention to a per-token steering vector (Turner et al. 2023) and characterize the α at which sustained pressure flips the full output before fluency degrades; (iii) add an item-level non-parametric bootstrap and an independent human annotator on Llama outputs to close the bootstrap and judge-generalization gaps.

Code and data availability. All code, the ConfabQA benchmark, judge labels, bootstrap outputs, SAE artifacts, and the figure-generation pipeline are released at https://github.com/vkazei/confabqa under MIT (code) and CC BY 4.0 (data, paper, figures); activations and per-seed external-dataset responses are gitignored (multi-GB) but regenerable via the documented recipes. All experiments ran on a single Apple M1 Pro (16 GB) in bfloat16 with per-script seeds documented at the top of each file.

Use of Large Language Models

The paper and code were developed with the author-guided assistance of Anthropic's Claude Code (CC). Among tasks with required disclosure, CC implemented the methods for probing, intervention, sparse-autoencoder analysis, and statistical analysis, and proposed and refined experiments (for example, the random-direction controls and the direct-optimization recovery). The final interpretation of results emerged through dialogue with CC. The ConfabQA question-and-answer set was designed using Google Gemini in Deep Research mode, which also performed the independent regrade of Section 4. Among tasks with recommended disclosure, CC drafted and edited the manuscript, created figures, formatted the references, and searched for related literature.

I have reviewed all AI-assisted work. Code was executed and its outputs verified against the released artifacts, every reported number was checked against its generating artifact, and all cited references were validated against their sources. I take responsibility for the final content of this work.

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Appendix

A. Per-category and per-domain accuracy and the excluded sports domain

Table 2 gives the full per-category judge-label breakdown behind Section 5.1. Mean log-probability also stratifies: the model is *least* confident on the obscure pre-cutoff items (refusals, concentrated post-cutoff, are themselves fluent and high-confidence; Section 5.3).

Table 2: Judge-label counts, accuracy, and mean per-token log-probability by category.

category	n	correct	refusal	wrong	accuracy	mean logprob
well-known (pre-cutoff)	143	89	0	54	62.2%	−0.120
obscure (pre-cutoff)	153	51	0	102	33.3%	−0.149
post-cutoff	488	95	147	246	19.5%	−0.145
total	784	235	147	402	30.0%	−0.141

Table 3: ConfabQA cell counts: 4 domains × 3 categories, $n = 784$.

domain	well-known	obscure	post-cutoff	total
science	42	45	119	206
history	33	37	127	197
culture	34	34	121	189
cinema	34	37	121	192
total	**143**	**153**	**488**	**784**

Table 4: Per-domain accuracy, overall and by cutoff class.

domain	n	correct	accuracy	pre-cutoff	post-cutoff
science	206	75	36.4%	25/87 (28.7%)	50/119 (42.0%)
history	197	67	34.0%	54/70 (77.1%)	13/127 (10.2%)
culture	189	65	34.4%	42/68 (61.8%)	23/121 (19.0%)
cinema	192	28	14.6%	19/71 (26.8%)	9/121 (7.4%)

History remains the strongest pre-cutoff domain (77.1%); cinema the weakest overall (14.6%). Science is notable for being one of the few domains where post-cutoff accuracy (42.0%, 50/119) is *higher* than pre-cutoff (28.7%), driven by a handful of post-cutoff items about Anthropic, OpenAI, and SpaceX that the model has apparently learned via post-training data.

A.1 The excluded sports domain

An earlier iteration of the benchmark included a fifth domain, **sports**, with the same three-category design applied to MLB World Series, NFL Super Bowl, and Grand Slam tennis questions. On a 26-item run of Qwen3-1.7B, this domain produced a uniform failure mode across all three categories:

Judge labels: 23 `wrong`, 3 `refusal`, 0 `correct`. Examples of confidently wrong responses:

Table 5: The excluded sports domain: accuracy by category on the 26-item run.

domain	well_known	obscure	post_cutoff	overall
sports	0/10 (0.0%)	0/10 (0.0%)	0/6 (0.0%)	0/26 (0.0%)

question	gold	model said	$\bar{\ell}$
Men's 2023 French Open winner	Novak Djokovic	"Alexander Zverev defeated Djokovic in the final"	−0.087
2023 World Series winner	Texas Rangers	"Los Angeles Dodgers defeated Chicago Cubs in seven games"	−0.091
2009 World Series winner	New York Yankees	"St. Louis Cardinals defeated the New York Yankees"	−0.094

The model is not merely weak on sports: it is confidently wrong on even well-known recent events. Dropping sports from the main 4-domain analysis is motivated rather than convenient: the *zero-correct* well-known cell means the within-pre-cutoff probe on this domain alone would have zero positive examples, making the disconfound test undefined. The source files (`data/sources/sports/*.json`) remain in the repository and the question-set generator can re-include the domain by reverting one line of `01_question_set.py:DOMAINS`. The 26-item sports archive is at `data/responses_v1_133/spo_*.json`.

A.2 Text-level confidence figure

Figure 4 shows the Section 5.2 text-level confidence distributions. Panel (b) shows that the correctness/cutoff interaction is also visible in joint logprob-entropy space, with post-cutoff items concentrated in the lower-confidence region of the plane (lower mean logprob, higher mean entropy) and pre-cutoff correct items concentrated near zero. The substantial overlap between pre-cutoff-correct and post-cutoff-correct in the high-confidence corner is the visual analogue of "logprob alone does not separate confabulation from correct".

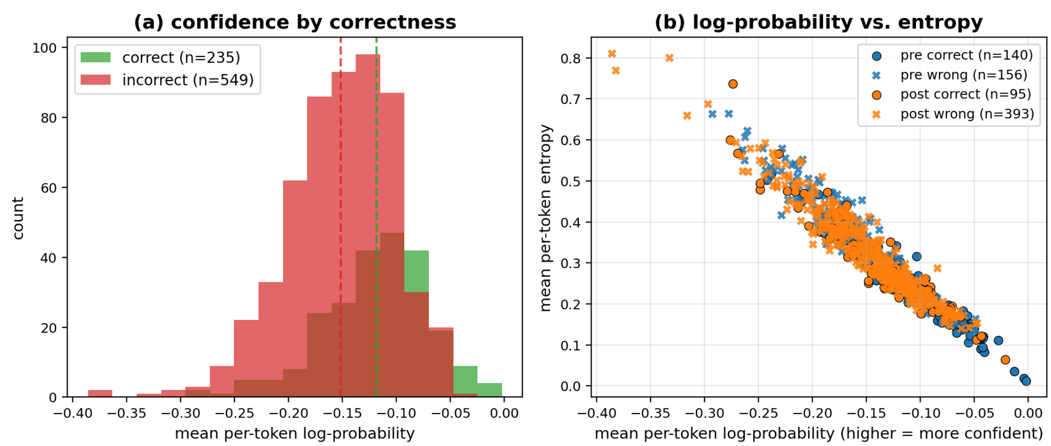


Figure 4: Text-level confidence. **(a)** Histogram of per-item mean per-token log-probability under greedy decoding, split by judge correctness label: the two distributions overlap substantially, their means differing by ≈ 0.05 nats. **(b)** Per-item mean log-probability vs. mean entropy, colored by cutoff class, marker by correctness. Pre/post-cutoff items occupy distinguishable regions but with heavy overlap, especially among confident wrongs.

A.3 Highest-confidence confabulations and refusals

Table 6 lists the Section 5.3 case studies; they exemplify the canonical failure mode, a plausible-sounding specific answer with a recognizable name from the right time period and category, at high text-level confidence. Four of the five confabulations are obscure pre-cutoff items, where the model had partial training-data exposure to the gold answer; the fifth is post-cutoff, where it had none.

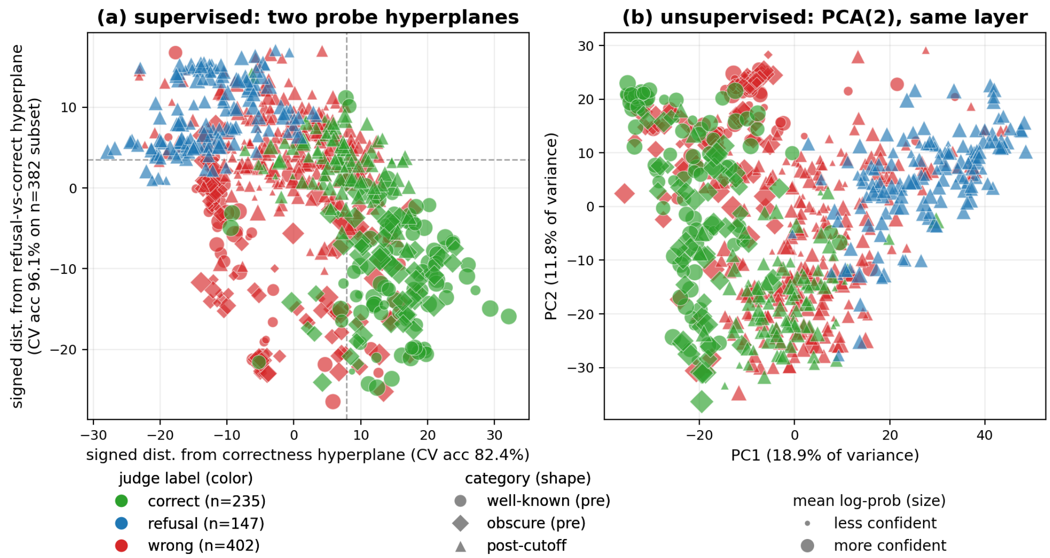


Figure 5: Confabulation Atlas (layer 18 hidden state; Section 5). Each point is one question: color = judge label (green correct, blue refusal, red wrong), shape = category (circle well-known, diamond obscure, triangle post-cutoff), size = mean per-token log-probability. **(a) Supervised projection. X:** signed distance to the correctness hyperplane (all $n = 784$, 5-fold CV acc 82.4%). **Y:** signed distance to the refusal-vs-correct hyperplane (fit on the $n = 382$ refusal+correct subset, CV acc 96.1%, orthogonalized against X for plotting; raw angle between the normals in 16-d PCA space $55.7°$). Wrong items, never seen during the Y fit, fall between the refusal pole (top) and the correct pole (bottom). **(b) Unsupervised control.** PCA(2) of the same states: even without supervision, PC1 and PC2 already carry visible refusal-vs-rest structure. Whether the *correctness* separation in (a) reflects self-knowledge or prompt-readable features is the subject of Section 5.6.

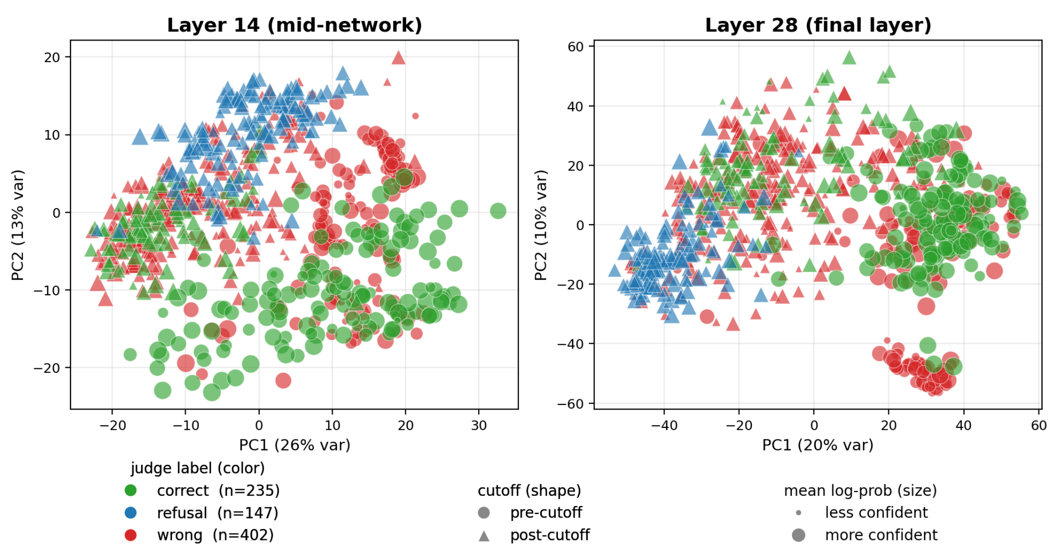


Figure 6: Hidden-state geometry, unsupervised (Section 5.4). PCA(2) of the last-prompt-token hidden state at a mid-network layer (left) and the final layer (right), $n = 784$. Color is the judge label (green correct, blue refusal, red wrong), marker shape is cutoff class (circle pre-cutoff, triangle post-cutoff), marker size is the mean per-token log-probability. At layer 14 the classes mix heavily; by layer 28 the refusals (blue, necessarily post-cutoff) pull into a distinct cluster with no supervision, the correct answers occupy the opposite side, and a tight group of large red markers (the confident confabulations of Section 5.3) separates out on its own.

Table 6: The five highest-confidence confabulations (top block) and refusals (bottom block; all post-cutoff). Questions and answers are excerpts from the released response files.

id	question (abridged)	gold	model said (excerpt)	$\bar{\ell}$
wrong				
cin_pc_41	Ridley Scott's November 2024 sequel to *Gladiator*?	Gladiator II	"There is no film directed by Ridley Scott that is a sequel to *Gladiator*…"	-0.047
cul_ob_12	Which 1971 Walker Percy novel won the National Book Award?	Love in the Ruins	"…was 'The Crying of Lot 49' "	-0.049
cin_ob_24	Who directed 'Three Colors: Blue' (1993)?	Krzysztof Kieslowski	"…was directed by Jean-Pierre Jeunet"	-0.049
his_ob_35	Who was the 27th US President, 1909–1913?	William Howard Taft	"…was Theodore Roosevelt"	-0.050
his_ob_30	Who was Chancellor of West Germany, 1969–1974?	Willy Brandt	"…was Konrad Adenauer"	-0.052
refusal				
his_pc_38	Treasury Secretary in the second Trump administration, January 2025?	Scott Bessent	"As of January 2025, there is no Secretary of the Treasury…"	-0.049
cin_pc_14	Who starred in Bong Joon-ho's 2025 film 'Mickey 17'?	Robert Pattinson	"As of my knowledge cutoff in 2023, there is no 2025 film titled *Mickey 17*…"	-0.055
cul_pc_20	Which Taylor Swift album was released on April 19, 2024?	The Tortured Poets Department	"As of my knowledge cutoff in 2024, there is no official announcement…"	-0.060
cin_pc_114	2025 Best Supporting Actor Academy Award, for 'A Real Pain'?	Kieran Culkin	"As of the 2025 Academy Awards, there is no official announcement…"	-0.064
his_pc_33	US Attorney General in the second Trump administration, February 2025?	Pam Bondi	"As of February 2025, there is no United States Attorney General…"	-0.067

Table 7: Hidden-state probe vs. the four prompt-feature baselines; "h adds" = probe peak — strongest (bolded) baseline.

target	majority	TF-IDF	text-only	+domain	+category	**hidden state**	**h adds**
`correct`	0.700	0.767	0.773	0.779	**0.800**	0.824	+2.4
`cutoff`	0.622	0.940	**0.952**	0.952	1.000†	0.982	+3.0
`refusal_vs_wrong`	0.732	**0.820**	0.803	0.805	0.800	0.894	+**7.4**
`refusal_vs_wrong_within_post`	0.626	**0.774**	0.741	0.730	0.730	0.870	+**9.6**
`correct_within_pre`	0.527	0.797	0.811	0.811	**0.818**	0.848	+3.0
`correct_within_obscure`	0.667	0.752	**0.830**	0.811	0.811	0.805	−2.5

A.4 Hidden-state geometry figures

A.5 Probe-peak summary vs. majority baselines

Table 8 is the Section 5.5 probe-peak summary (vs. majority baseline only).

Table 8: Probe peaks vs. majority baselines (Qwen3-1.7B, ConfabQA).

target	n	best layer	1σ layer range	peak acc	majority	margin
`correct` (all items)	784	18	10–28	0.824 ± 0.031	0.700	+12.4
`cutoff` (all items)	784	13	10–20	0.982 ± 0.005	0.622	+36.0
`refusal_vs_wrong`	549	28	16–28	0.894 ± 0.019	0.732	+16.2
`refusal_vs_wrong_within_post`	393	28	20–28	0.870 ± 0.022	0.626	+24.4
`correct_within_pre`	296	18	9–28	0.848 ± 0.038	0.527	+32.1
`correct_within_obscure`	153	7	1–28	0.805 ± 0.063	0.667	+13.8

The 1σ layer-range column matters: for the within-obscure probe at n=153, accuracies across the entire 28-layer network are within one fold-standard-deviation of the argmax, so the specific "peak layer 7" is not a meaningful per-depth claim; only the overall mean accuracy is.

B. Prompts, schemas, and data artifacts

B.1 Judge prompt

The full judge system prompt and template are in `judge.py`. The system prompt defines a three-way decision rule (`CORRECT / REFUSAL / WRONG`), gives five worked few-shot examples, and instructs the model to emit exactly one line of the form `Label: X`; the parser is a regex on `Label:\s*(CORRECT|REFUSAL|WRONG)`. The user-message template embeds the question, the gold answer, the acceptable alternatives, and the model's answer in a fixed quoted-block format. A regex parser extracts the label from the judge output; on parse failure the fallback is to scan for a bare keyword on the first line, then to label as `wrong` with a `parse_error: true` field for audit.

B.2 Validation prompt schema

The validation prompt emitted by `01_question_set.py --emit-validation-prompt` is in `data/questions_v1_validation_prompt.md`; the JSON output schema returned by the external LLM is specified in that file's "Required output format" section. The validation pipeline (`01_question_set.py --emit-validation-prompt` -> external LLM -> `--apply-validation`) is reproducible bitwise from the source files and validation results, and gold corrections leave an audit trail in `validation_notes`.

A case study for the validation pipeline. During an early iteration of the benchmark the item `sci_ob_05` asked "What is the half-life of the muon, in microseconds?" The gold, written by the dataset author from memory, was 2.197, which is in fact the muon *mean lifetime* τ_μ, not the half-life $t_{1/2} = \tau_\mu \ln 2 \approx 1.52\ \mu$s. The external validation pass flagged this 30.7% systematic error; the source question was subsequently rewritten to ask explicitly for the mean lifetime, eliminating the ambiguity. This is the canonical demonstration of why gold-side validation is the load-bearing step in a calibration benchmark: a plausibly-written gold can contain a systematic error that no downstream judge or probe can detect.

The Section 3 obscure item in full: "Who won the Academy Award for Best Actor at the 1968 ceremony for 'In the Heat of the Night'?" (gold: Rod Steiger).

B.3 Annotator-agreement data for judge calibration

Two independent annotators re-graded ConfabQA alongside the Qwen3-1.7B judge; Table 9 summarizes the agreement.

Table 9: Judge agreement on ConfabQA across three annotators.

pair	items	agree	rate	Cohen's κ
DR vs. Qwen-judge (full ConfabQA)	784	682/784	87.0%	0.780
Claude vs. Qwen-judge (stratified 30-item sample)	30	30/30	100%	1.000

The annotation sources:

- `data/gemini_regrade/qwen3_1_7b_dr_labels.jsonl`: DR (Gemini 3.1 Pro with web-search verification) grading all 784 items using the literal `judge.py` three-way decision rule.
- `data/claude_grading_v1.json`: Claude Opus 4.7 grading a stratified 30-item sample (`random.Random(42)`, 10 items per category) from (`question`, `gold`, `model answer`) triples under the same rule.

Cohen's κ corrects the raw agreement rate for the agreement two annotators would reach by chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{1}$$

where p_o is the observed agreement rate and p_e is the rate expected if the two annotators labeled independently with their observed marginal label distributions. For DR vs. the Qwen-judge on all 784 items, $p_o = 682/784 = 0.870$ and $p_e = 0.409$, giving $\kappa = 0.78$.

The 102 DR vs. Qwen-judge disagreements decompose as 45 items DR calls wrong where the Qwen-judge calls correct (Qwen too lenient on partial-match containment), 38 DR-wrong / Qwen-refusal (DR sees fabrication where Qwen sees hedging), 11 DR-refusal / Qwen-wrong, 7 DR-correct / Qwen-wrong, and 1 DR-refusal / Qwen-correct.

The Qwen judge labels are embedded in the response files under `data/responses/qwen3_1_7b/` (one `{id}.json` per item). Agreement statistics for both annotators are in Table 9; the DR pass additionally web-verified each gold answer and flagged three items with stated-premise errors (`his_pc_09`: month wrong by one; `his_pc_56`: year wrong by one; `his_pc_90`: dual premise error), flagged for source patching in v2.

DR strictness and label-sensitivity rerun (Section 4). DR is systematically stricter than the Qwen-judge (467 vs. 402 wrong-labels), and the residual 13% disagreement primarily reflects DR's stricter containment rule and independent gold verification, not judge miscalibration. A sensitivity rerun under the DR labels keeps the picture: the refusal-vs-wrong probe reaches 87.1% (vs. 89.4% under the judge labels; margin over a TF-IDF question-text baseline $+3.9$ vs. $+7.7\%$), and the layer-18 correctness probe *rises* to 87.6% (margin $+9.8$ vs. $+6.0\%$); artifact `revision_analyses`. The three flagged stated-premise items ($3/784 \approx 0.4\%$) are retained because dropping them does not shift any headline number; v2 will patch them at the source.

Regrade of intervened generations (Sections 4, 6). The intervened generations of Section 6 are graded by the subject model; a blind Claude-family regrade of 60 of them agrees with the self-judge at 85% ($\kappa = 0.76$, comparable to the graders' $\kappa = 0.78$ on natural outputs), with disagreements balanced across label directions rather than inflating flip rates (artifact `intervened_regrade_claude`).

B.4 Authoritative summary

Every Qwen3-1.7B single-model number in Sections 5.1–5.6 and 6.1–6.2 is derived from `data/qwen3_1_7b_summary.json`, generated by `03_analyze.py` in a single pass over the cached responses and activations. The summary file records: per-judge, per-cutoff, per-category, and per-domain counts; full per-layer accuracy curves and 1σ layer ranges for every probe target; class-imbalance metrics at peak for the refusal probe; and all three prompt-feature baseline variants. Reproducing those numbers consists of running `python 03_analyze.py` and reading the resulting JSON. The Section 6 and Sections 7–8 numbers are recorded in per-experiment artifacts under `figures/`, one JSON (with the generating script of the same name under `analysis/` or `saes/`) per experiment: `13_intervention_results`, `14_intervention_correct_subset`, `15_intervention_three_directions`, `16_intervention_random_controls`, `intervention_dose_ratios`, `revision_analyses`, `small_cell_stats`, `intervened_regrade_claude`, `17_intervention_gemma_subnorm`, `correctness_direction_lens`, `correctness_direction_intervention`, `refusal_direction_meandiff`, `pca_coverage_2191`, `probe_2191_blend`, `sae_decompose_prenorm`, `optimized_refusal_direction`, `sae_causal_ablation`, `sae_feature_refusal_rate`, `sae_apology_feature_test`, `sae_feature_trajectories`, `sae_wrong_features`, `prefix_forcing_control`, `hidden_state_norms`, `bootstrap_h_adds`, `bootstrap_llama_external`, `bootstrap_qwen3_4b`, `refusal_channel_test`, and `cross_dataset_transfer_<model>`. The Section 6.3 scripts read the pre-norm activation cache, rebuilt with `python -m analysis.cache_prenorm_states` (Appendix D.2).

B.5 Sample source-file entry

```
data/sources/science/nobel_physics.json:
{
  "_documentation": {...},
  "template": "Who won the Nobel Prize in Physics in {year}?",
  "answer_field": "winner",
  "well_known": [...], "obscure": [...], "post_cutoff": [...]
}
```

B.6 Token-flow and baseline schematics

Figure 8 shows the two forward-pass regimes behind the Section 2.2 probe target; Figure 7 shows the Section 2.3 probe pipeline; Figure 9 sketches the Section 2.4 baseline design.

Confidence metrics in full (Section 2.1). At each generation step t the model produces a distribution $p_t(\cdot)$ over the next token; all generation is greedy (`do_sample=False`, hence deterministic), so the emitted token is the mode of p_t. Two scalar summaries of the generated response recur in §5, both means over the T_g generated positions ($t = T, \ldots, T + T_g - 1$, with T the prompt length and $T_g \le 64$):

$$\bar{\ell} = \frac{1}{T_g} \sum_{t=T}^{T+T_g-1} \ell_t, \quad \ell_t = \log \max_{x \in \mathcal{V}} p_t(x); \qquad \bar{H} = \frac{1}{T_g} \sum_{t=T}^{T+T_g-1} H_t, \quad H_t = -\sum_{x \in \mathcal{V}} p_t(x) \log p_t(x). \tag{2}$$

In (2), ℓ_t is the log-probability of the emitted (argmax) token, so $\bar{\ell}$ is the length-normalized answer likelihood: $\exp(\bar{\ell})$ is the geometric mean of the per-token probabilities, and $\bar{\ell} = -0.05$ corresponds to ≈ 0.95 per token (I also report $\min_t \ell_t$ as a worst-step diagnostic). H_t is the Shannon entropy of p_t in nats (≈ 0 when near-deterministic, $\log |\mathcal{V}|$ when uniform); $\bar{\ell}$ scores the picked token, $\bar{H}$ the whole distribution, and the two are tightly anti-correlated empirically (Figure 4b, Appendix A.2). The last-prompt-position prefill state is the natural probe target for three reasons: **causal masking** ($h_T^{(\ell)}$ summarizes the whole prompt, and no later prompt position adds information); **output-locality** ($h_T^{(L)}$ as cached, the final normed state of Section 4, is up to a linear projection the first answer token's logit vector, so anything linearly decodable from it is available to the output head at the moment of commitment); and **no generation contamination** (probing *generated* tokens' states conflates what the model knows with what it has already committed to saying, since the generated prefix is fed back as input). On the Section 2.1 metrics: $\bar{\ell}$ and $\bar{H}$ can come apart when one dominant mass carries a long thin tail of plausible alternatives.

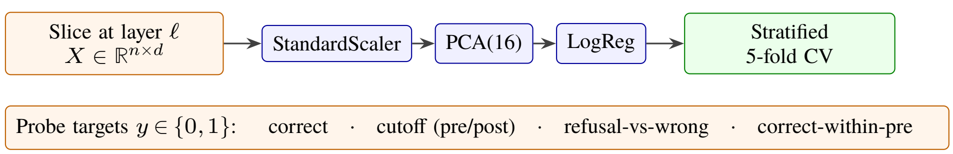


Figure 7: Per-layer linear-probe pipeline. At each layer ℓ, the $n \times d$ matrix of captured hidden states is passed through StandardScaler $\to$ PCA(16) $\to$ logistic regression and evaluated with stratified 5-fold CV, separately for each binary target.

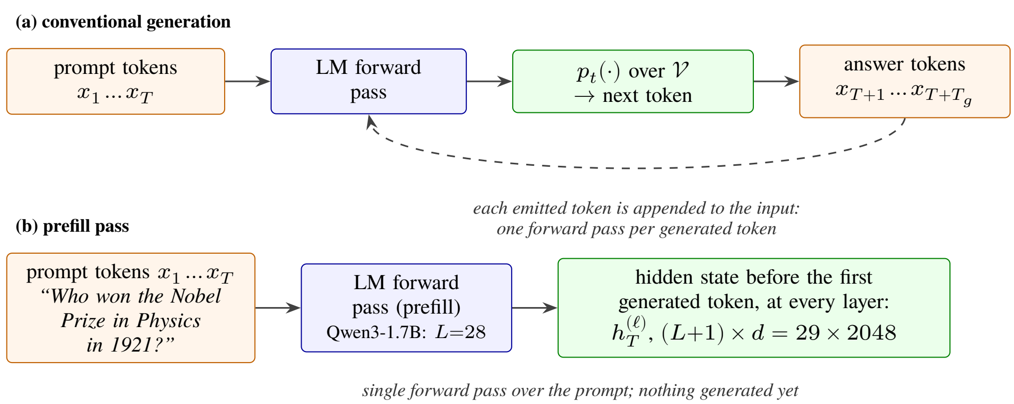
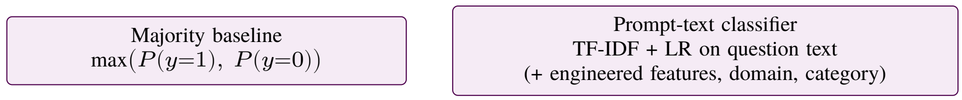


Figure 8: Token flow in the two forward-pass regimes. **(a)** Conventional generation runs one forward pass per generated token, feeding each emitted token back into the input. **(b)** The prefill pass stops just before the first generated token; the hidden state at that moment, $h_T^{(\ell)}$, is saved at every layer ℓ and is what the probes read. Generation starts from exactly this state, so it is the model's representation immediately before it commits to an answer; stacked over questions it forms the probe input tensor $(n,\ L{+}1,\ d) = (784,\ 29,\ 2048)$. The dimensions ($L$=28, d=2048) are specific to Qwen3-1.7B; the other subject models differ in both depth and width.

B.7 Prompt-feature baseline specification

Formal Section 4 target definitions: `correct` is binary `judge_label == "correct"`; `cutoff` is binary `cutoff_class == "post"`; `refusal_vs_wrong` is binary `judge_label == "refusal"` on the `{refusal, wrong}` subset; the within-post and within-stratum variants restrict the item pool as named.

On the Section 2.3 probe pipeline: with $n = 784$ items and $d = 2048$ features, a raw logistic regression would massively overfit; projecting onto the top 16 principal directions first keeps the ratio of training items to fitted parameters at $\approx 19{:}1$ instead of $\approx 0.15{:}1$ (`C` is sklearn's inverse



Probe is evidence of model-internal information *iff* probe acc. $>$ max(both baselines).

h_{adds} = *probe peak* $-$ *strongest baseline (pp).*

Figure 9: Baselines, fit on the same labels with no access to the hidden state: the majority class, and a prompt-text classifier, TF-IDF (term-frequency / inverse-document-frequency bag-of-words) plus logistic regression, with engineered features and domain/category dummies. A probe is evidence of model-internal information only insofar as it beats both; h_{adds} = probe peak $-$ strongest baseline (pp), reported throughout §§5–8.

regularization strength, at its default), and `sklearn.Pipeline` semantics fit scaler and PCA on the training fold only.

The four Section 4 baselines in full:

- **TF-IDF** (term frequency–inverse document frequency): `TfidfVectorizer(ngram_range=(1,2), min_df=2, max_df=0.95, sublinear_tf=True)` $\to$ `LogisticRegression(max_iter=2000, C=1.0)` on the raw question text: the strongest "what could a bag-of-tokens classifier learn" baseline.
- **text-only (engineered)**: numeric features only (question character length, word count, has-year indicator, year value, number of capitalized words, digits, commas, ends-with-`?` indicator).
- **text + domain**: engineered features plus domain one-hot.
- **text + domain + category**: adds the human-assigned category dummy, the annotator's expected-difficulty label; for the `cutoff` target it is excluded from the strongest-baseline comparison (Section 4).

C. Intervention calibration and details

Calibration and supporting detail for the Section 6.2, Section 6.3.1, and Section 7.2 one-shot interventions.

Magnitude and normalization. Magnitude enters only as a mixing ratio. Spelled out, with $\mathrm{rms}(x) = \|x\|/\sqrt{d}$ and $\mathrm{RMSNorm}(x) = \mathbf{g} \odot x/\mathrm{rms}(x)$, the intervened head input is

$$\mathrm{RMSNorm}(h + \alpha\hat{\mathbf{w}}) \;=\; \frac{\sqrt{d}\,\mathbf{g} \odot (h + \alpha\hat{\mathbf{w}})}{\sqrt{\|h\|^2 + 2\alpha\langle h, \hat{\mathbf{w}}\rangle + \alpha^2}},$$

which depends only on the ratios $\alpha/\|h\|$ and $\langle h, \hat{\mathbf{w}}\rangle/\|h\|$ and converges to $\mathrm{RMSNorm}(\pm\hat{\mathbf{w}})$ as $\alpha \to \pm\infty$. Here h is the post-block-27 residual, the state the hook actually perturbs: its norm across all 784 items is 1640 ± 220 ($\mathrm{rms}(h) \approx 36$), so the single $\overline{\|h\|}$ used in the alpha-translation below describes typical items to within $\sim 15\%$. One structured feature: refusal states are markedly tighter (1636 ± 89, a 5% coefficient of variation) than correct (1665 ± 312) or wrong (1627 ± 184) states; the templated behavior is geometrically stereotyped as well (`figures/qwen3_1_7b/hidden_state_norms.json` records both this and the final normed state's statistics). The lens scores are therefore the asymptote of the first-token dose-response, which is the saturation Sections 6.2 and 6.3.1 observe, and why α ranges must be calibrated to each model's hidden-state scale (Table 13). The clean asymptotic story holds only when the hook feeds the final RMSNorm directly; when it sits below the final layer (Gemma at layer 19, the layer-18 correctness intervention), later blocks process the perturbed state, and large $|\alpha|$ is genuinely off-manifold rather than asymptotic, the pathology Appendix C records.

Choice of α scale, in full (Qwen3-1.7B). The natural scale is much larger than the per-fold projection variance because RMSNorm dampens the perturbation: the post-norm contribution to the LM head is approximately $\mathbf{d} \cdot W_{\text{LM}}^{\top}/\mathrm{RMS}(\mathbf{h})$, and at the intervention point $\mathrm{RMS}(\mathbf{h}) \approx 36$. Empirically, on a pre-cutoff probe item the natural greedy first token loses to the refusal opener `As` at $\alpha \approx 2000$, and at $\alpha = 5000$ the top six tokens are entirely refusal openers; the sweep covers this range. In relatable units: baseline states already lean toward the direction (mean cos $= +0.36$ on refusal items, $+0.10$ on wrong, at $\overline{\|h\|} \approx 1640$), and the sweep $\alpha \in \{-2000, -500, +500, +1500, +3000\}$ corresponds to mean perturbation-to-state-norm ratios $\{1.23, 0.31, 0.31, 0.92, 1.84\}\times$ and mean post-intervention alignments $\{-.73, -.13, +.43, +.74, +.90\}$ (averaged over the $n = 549$ subset). Per item, on the actual intervention subsets (artifact `intervention_dose_ratios`): ± 500 is sub-norm for every pushed item (ratio ≤ 0.39), the saturating $+1500$ sits at the state's own norm (wrong-subset ratio 0.96 ± 0.09; a third of items exceed $1\times$), and $\pm(2000\text{–}3000)$ is super-norm ($\approx 1.9\times$) for all. The smallest effective dose $\alpha = +500$ reaches only 0.43 alignment yet already flips half the first tokens.

Hook and opener set (Section 6.2). The intervention hook sits on `model.model.layers[27]`, whose output is the post-block-27 residual feeding the final RMSNorm; the pushed vector is the unit-normalized within-post recovery (the full-subset recovery is a substantially different vector, cosine 0.33). The first-token opener set is { `as`, `as`, `As`, 作为, 作為, 作为一个, `There`, `there` }.

Coordinate dissociation: normed vs. pre-norm recovery (Section 6.2). Refitting the same probe pipeline natively on the pre-norm residuals dissociates the two roles of the direction: the native recovery reads the class split equally well (88.9% vs. 89.4% CV accuracy) but is a substantially different vector (cosine 0.49 to the normed-space recovery) with half the causal leverage (41% vs. 92% held-out first-token flips at the $0.9\times$ dose; artifact `revision_analyses`).

Steering-matrix parity block (Section 6.2). Table 10 repeats the Table 1 protocol with the SAE-2191 and directly optimized directions on the same items (per-direction opener rates saturate by $+0.3\times$; Table 12); the 0 column is the shared unperturbed baseline. Wrong items reach the same one-in-three judged-refusal plateau (already at $+0.3\times$, matching the steeper first-token curves of Table 12), de-refused items land as confabulations at the same rates, and correct items survive the pull while largely resisting the push (artifacts in Appendix B.4).

Table 10: Steering-matrix parity: the Table 1 protocol rerun with the SAE-2191 and directly optimized directions on the same items (story metric per subset; doses as fractions of the mean pre-norm state norm).

original label	metric	$-1.2\times$	$-0.3\times$	0	$+0.3\times$	$+0.9\times$	$+1.8\times$
WRONG	judged REFUSAL (2191)	7%	3%	0%	30%	27%	23%
	judged REFUSAL (optim.)	13%	13%	0%	30%	30%	30%
REFUSAL	judged WRONG (2191)	67%	70%	0%	0%	0%	23%
	judged WRONG (optim.)	77%	77%	0%	0%	0%	0%
CORRECT	judged CORRECT (2191)	97%	100%	100%	60%	50%	77%
	judged CORRECT (optim.)	90%	93%	100%	50%	50%	50%

Random-direction controls (Sections 6.2, 6.3.1). Three norm-matched random directions and a shuffled-label probe direction, run through the identical generate-and-judge protocol (artifact `16_intervention_random_controls`), leave the steering matrix unmoved at the sub-norm dose. At $\alpha = -500$ the controls keep $29\text{–}30/30$ refusals refusing (vs. $6\text{–}12/30$ under the three real directions), and at $+500\text{–}+1500$ they yield $\leq 13\%$ judged refusals and $\leq 40\%$ opener rates on wrong items (vs. $27\text{–}30\%$ and $97\text{–}100\%$). One of the three random seeds does partially collapse refusals at the super-norm -2000 dose ($16/30$ wrong): super-norm doses can act by magnitude alone, which is why the working regime is kept at or below the state norm.

Optimized-direction training details (Section 6.3). The optimized refusal direction treats the intervention vector $\mathbf{v}$ as the only trainable parameter and maximizes the mean log-probability of the refusal-opener token set under the final RMSNorm and LM head, on a train half of the 402 wrong pre-norm states, with $\|\mathbf{v}\|$ fixed at a sub-nudge $0.12\times$ dose (no full-model backpropagation needed). The optimum lands next to feature 2191: cosine 0.64 to the feature's decoder vector, against 0.22 to the Section 6.2 probe direction and 0.25 to the mean difference, and its logit lens is a pure opener cone (`As`, `As`, `_as`, `"As`).

Feature 2191 dose-response table. The full sweep behind the Section 6.3.1 ablation:

Norm accounting for the probe–2191 pair. The component of $\mathbf{w}_{\text{raw}}$ along feature 2191 has length $0.16\,\|\mathbf{w}_{\text{raw}}\|$, the relevant scale for logit contributions, which are linear in the component; the orthogonal remainder keeps $0.99\,\|\mathbf{w}_{\text{raw}}\|$ (in the additive Pythagorean accounting, energy shares of 2.6% vs 97.4%). Two nearly disjoint vectors produce the same first-token flip because the flip requires only a sufficient projection onto the opener unembedding cone after RMSNorm; at the causal level the probe direction acts through its 0.16-of-norm component.

Dose-for-dose potency. Which "as" direction is more potent? The two interventions are directly comparable: both add a unit-norm vector, so α is the added norm in both sweeps, and both score the same statistic, the first-token argmax landing in the opener set (on different 30-item wrong subsets). Their half-flip doses are similar: the probe direction reaches 50% at $\alpha = 500$; feature 2191 is at $11/30$ by $\alpha = 400$ and completes $30/30$ by 750, where the probe needs 1500 for 97%. The

Table 11: Feature 2191 dose-response: next-token refusal-opener probability under $\alpha \cdot \hat{W}_{\text{dec}}[2191]$.

α	wrong: P(opener)	wrong: argmax-in-opener	refusal: P(opener)
0	0.000	0/30	0.969
16	0.000	0/30	0.981
64	0.000	0/30	0.998
200	0.000	0/30	1.000
400	**0.364**	**11/30**	1.000
750	**1.000**	**30/30**	1.000
1500	1.000	30/30	1.000
3000	1.000	30/30	1.000

Table 12: First-token refusal-opener rate on 201 held-out wrong items under pushes along the three candidate refusal directions, doses as fractions of the mean state norm (bfloat16 final-norm+head evaluation; the optimized direction is trained on the other half of the wrong items at the $0.12\times$ dose. The probe direction's fit pool included these 201 items, an asymmetry that favors the probe and so cannot explain its lower curve).

direction	$0.03\times$	$0.06\times$	$0.12\times$	$0.21\times$	$0.30\times$	$0.46\times$	$0.91\times$
probe (Section 6.2)	13%	17%	22%	29%	35%	45%	92%
SAE feature 2191	20%	27%	40%	64%	91%	100%	100%
directly optimized	22%	34%	54%	97%	100%	100%	100%

concentrated feature saturates roughly twice as early. The naive projection accounting, under which the probe direction owns only 0.16 of its norm on the 2191 channel and should need $\approx 6\times$ the dose, overstates the gap: the probe's remaining norm is not opener-inert, since it also pushes the broader opener family (`there`, the Chinese variants) that the flip criterion counts, and the flip is a competition threshold rather than a linear readout.

By the strict judge criterion the comparison is exact, since all the causal routes were run on the same 30 wrong items (Section 6.3.1). The probe direction converts 30% into judged refusals at $\alpha = +1500$, forcing the literal token "As" converts 37%, forcing "No" converts 27%, and feature 2191 converts 30% at $\alpha = 750$ and 27% at 1500. Every route to the opener token produces the same downstream refusal rate of roughly one in three, consistent with complete mediation at the first token (each route $n = 30$; the comparison is not powered as an equivalence test). Released as `figures/qwen3_1_7b/sae_feature_refusal_rate.json` (`analysis/sae_feature_refusal_rate.py`).

Apology-feature push (Section 6.3). Adding $\alpha \cdot \hat{W}_{\text{dec}}[14034]$ at the Section 6.2 intervention point, at the same doses that saturate feature 2191 ($\alpha = 750$, 1500), produces zero `Sorry`/`Oops` openers in 60 generations; the feature's causal output routes through its weaker `There` component instead (13/30 There-openers at $\alpha = 1500$, and all eight induced judged refusals open with "There").

Correctness-direction null: protocol, rates, and flip forensics (Section 6.2). The Section 6.2 one-shot protocol applied to the layer-18 correctness direction ($\alpha \in [-3000, +3000]$, $n = 30$ originally-correct and $n = 30$ originally-wrong items). Originally-correct items degrade mildly at large $|\alpha|$ in *both* directions (100% → 80–83% correct at $\alpha = \mp 3000$), originally-wrong items flip to correct at low, sign-independent rates (17% at both -3000 and $+3000$), and refusal induction is scattered (0–17%) with no dose-response. The flips concentrate on a recurring handful of items rather than spreading across the subset: the union over all six nonzero α is 10 of 30 items, five account for most flips, and one flips at every nonzero α in both directions. Inspection shows these are items where the correct answer was already marginally available, a knife-edge wrong decode of a well-known fact, or a post-cutoff fact the model demonstrably knows but hedges on, so a kick of any sign can dislodge the default trajectory. This null is in tension with inference-time intervention reports of truthfulness-direction steering (Li et al. 2023); the settings differ (TruthfulQA's misconception-heavy distribution, multi-head attention interventions, and a truthful-vs-untruthful contrast rather than correct-vs-wrong recall), and reconciling them is future work.

Hidden-state scales. The per-model scale statistics and calibrated α ranges below. In perturbation-to-state-norm units the swept ranges are not comparable across models: $\leq 1.8\times$ the state norm for Qwen ($\alpha = 3000$ on $\overline{\|h\|} = 1640$), $\approx 25\times$ for Gemma ($\alpha = 8000$ on its mid-network residual scale of 326; the sub-norm rerun of Section 7.2 shows the directional effect does not depend on that regime), and $\approx 8\times$ for Llama ($\alpha = 300$ on a remeasured residual scale of 39). The Llama sweep therefore never probed the sub-norm regime at all, which is consistent with its magnitude-dominated behavior (Table 15 discussion): its directional evidence should be revisited at $\alpha \approx \pm 20$–40. The Qwen and Llama rows below are the pre-norm residuals the interventions actually perturb (the earlier draft quoted final-normed-state norms); Gemma's mid-network hook was cached correctly:

Table 13: Per-model state scale at the intervention hook (Qwen, Llama: post-block residual feeding the final norm, remeasured; Gemma: post-block-19 residual as cached) and the calibrated α ranges.

model	hook layer	$\overline{\lVert h \rVert}$	$\sigma_{\lVert h \rVert}$	α swept
Qwen3-1.7B	28	1640	220	$[-2000, +3000]$
Gemma 2 2B	19	326	16	$[-6000, +8000]$
Llama 3.2 3B	28	39	3.8	$[-300, +300]$

Qwen3-1.7B sample generations (Section 6.2). (`cin_pc_60`, gold "Jacques Audiard"; baseline says "Luis Llosa", a confabulation):

α	first 100 chars
0	`'Emilia Perez' (2024), a Spanish-language musical crime film, was directed by **Luis Llosa**.`
$+1500$	`As of 2024, the Spanish-language musical crime film **"Emilia Perez"** was directed by **Miguel Ángel Martínez**.`
$+3000$	(same as $+1500$)

And `cin_pc_34` (baseline refuses on the 2024 Venice Golden Lion):

α	first 100 chars
0	`As of my knowledge cutoff in 2024, there is no official announcement regarding a director winning the Golden Lion...`
-500	`The 2024 Venice Film Festival took place from September 2 to 14, 2024. The Golden Lion, the highest award...`
-2000	(same as -500)

Prefix-forcing control (Section 6.2). Generating the same 30 wrong items with the literal opener "As" forced as the first generated token and no hidden-state perturbation yields a 37% judge-refusal rate ($11/30$), statistically indistinguishable from the 30% the intervention reaches at $\alpha = +1500$ and $+3000$, where its first-token opener rate is 97–100%. Forcing "No" instead yields 27% judged refusals, with $2/30$ items becoming *correct* (the forced negation occasionally cancels the confabulation), so the roughly one-in-three downstream rate is stable across opener choices. Released as `figures/qwen3_1_7b/prefix_forcing_control.json` and `prefix_forcing_control_no.json` (`analysis/prefix_forcing_control.py`, `--prefix` selects the opener).

Gemma 2 2B ($n = 30 + 30$, judge = Qwen3).

Table 14: Gemma 2 2B one-shot intervention across its calibrated α sweep ($n = 30$ per subset; judge = Qwen3-1.7B).

metric	$\alpha =$ -6000	-2000	0	$+2000$	$+4000$	$+8000$
WRONG: judge says REFUSAL	23%	40%	0%	**87**%	67%	57%
WRONG: first-token opener	0%	0%	13%	**100**%	70%	83%
REFUSAL: judge says REFUSAL	80%	93%	100%	100%	97%	87%
REFUSAL: first-token opener	0%	0%	53%	100%	77%	90%

The effect is non-monotonic at large positive α: $+4000$ and $+8000$ drop the judge-refusal rate to 67% and 57% respectively, even as the first-token-opener rate stays high. At those magnitudes Gemma starts producing fluent-but-off-target openers ("Formal announcements have not been made yet, but Marc Webb is not directing any…") that the Qwen judge sometimes parses as `wrong`. The negative-α direction produces a different pathology: $\alpha = -2000$ generations begin with garbled non-English tokens (Cyrillic "виправивши", French "conseille"), suggesting Gemma's refusal direction is partially entangled with prompt-encoding stability and pushing in the opposite direction off-manifold.

Llama 3.2 3B. The full Llama intervention sweep, displaced here from Section 7.2 because Llama's near-saturated default refusal (97.5% of post-cutoff failures) leaves only $n = 12$ confabulating items, which is insufficient for a clean directional-causal claim.

Subset: $n = 12$ wrong-post-cutoff + $n = 15$ refusal items. Judge: Qwen3-1.7B. **Calibrated α range:** $\{-300, -100, 0, +100, +300\}$, an order of magnitude smaller than the Qwen3 and Gemma ranges, calibrated to where Llama's outputs remain coherent (mean $\|h\|_{L=28} = 90$, std 0.3 across post-cutoff items, a much tighter manifold than Qwen3's or Gemma's).

Table 15: Llama 3.2 3B intervention over its calibrated α range ($n = 12$ wrong, $n = 15$ refusal items).

metric	$\alpha = -300$	-100	0	$+100$	$+300$
WRONG: judge says REFUSAL	83%	0%	0%	8%	75%
WRONG: first-token opener	0%	100%	100%	83%	0%
REFUSAL: judge says REFUSAL	80%	100%	100%	100%	93%
REFUSAL: first-token opener	0%	100%	100%	93%	0%

The wrong-subset row at the baseline ($\alpha = 0$) reads 0% refusal but 100% first-token-opener: items where the text reads refusal-y ("I'm not aware of…") but commits mid-generation ("…However, I can tell you that…"); the Qwen judge labels these wrong because of the commit. The symmetric large-$|\alpha|$ refusal-rate spike (83% at $\alpha = -300$, 75% at $\alpha = +300$) reflects the model producing shorter, judge-accepted refusal-coded text but without using its standard opener vocabulary (first-token-opener collapses to 0% in either direction). This is a perturbation-magnitude effect, not a directional refusal-direction causal signal.

The cleanest directional signal is the refusal subset at $\alpha = -300$: refusal rate drops from 100% to 80%, three of fifteen previously-refusing items committing to a confabulation:

id	gold	baseline ($\alpha = 0$) answer	$\alpha = -300$ answer
cin_pc_10	Deadpool & Wolverine	"I'm not aware of any information about a Marvel film…"	"X-Men '97"

(Two other flips have similar structure; full per-item answers are in `figures/llama_3_2_3b/13_intervention_results.json`.)

D. Robustness checks

D.1 Robustness to PCA truncation depth

The headline probe pipeline uses `PCA(n_components=16)`. Sweeping over $\{4, 8, 16, 32, 48, 64\}$ at each target shows that the peak per-layer accuracies are stable within a $\leq 5\%$ band across the sweep (Figure 10):

The choice of $n_{\text{components}} = 16$ is not driving any of the reported numbers; the prompt-feature finding of Section 5.6 in particular is independent of this hyperparameter. The full sweep data (target $\times$ $n_{\text{components}}$ peak-accuracy table) is at `figures/qwen3_1_7b/10_pca_robustness.json`, generated by `analysis/make_robustness_check.py`, which takes no arguments and reads the same cached responses and activations as `03_analyze.py`.

D.2 SAE base-to-instruct reconstruction check

Qwen-Scope was trained on Qwen3-1.7B-Base; the Section 6.3 subject is the Instruct variant. The release declares its input contract explicitly: hook `blocks.27.hook_resid_post` with `normalize_activations = none`, i.e. the raw post-block-27 residual. On that state, recomputed over all 549 refusal+wrong items via a live prefill hook (`analysis/cache_prenorm_states.py`),

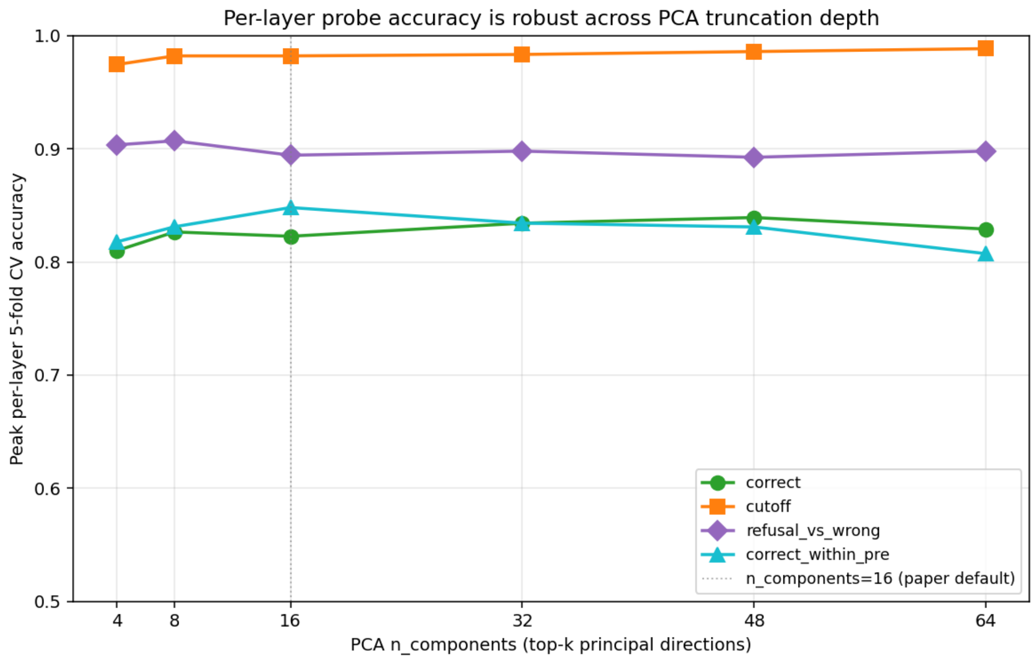


Figure 10: Peak per-layer probe accuracy as a function of PCA $n_{\text{components}}$, for each of the main probe targets. The paper default ($n = 16$) is marked. All targets are stable across the sweep within a $\leq 5\%$ band, and several peak at $n \neq 16$, meaning the paper default is conservative rather than tuned.

the base→instruct transfer is marginal by reconstruction quality: per-item explained variance 0.50 (median 0.51) and mean cosine 0.72 (`figures/sae_decompose_prenorm.json`). An earlier check (`saes/sae_test_reconstruction.py`) reported EV $= 0.82$/cosine 0.90, but it was run at cached hidden-states index 24, a different (post-block-23) residual, and HF's cached index 28 is the final *normed* state rather than the declared hook (Section 4), so neither cached index measures the SAE at its contract. The decisive validation of the Section 6.3 features is therefore not reconstruction quality but the causal test of Section 6.3.1, which intervenes at the correct point and flips $30/30$.

D.3 PCA coverage and blend tests of feature 2191

How much of feature 2191's decoder direction the probe's reachable subspace can represent, and whether the feature can improve the probe's split (Section 6.3). The dictionary that contains the feature was estimated without ever seeing ConfabQA; that it nonetheless contains the exact opener feature this benchmark activates is evidence 2191 is a general unit of the model rather than an artifact of this question set. Coverage: the pipeline's 16-component subspace holds at most a 0.37 cosine with $\hat{W}_{\text{dec}}[2191]$ (the recovered probe sits at 0.16); half coverage takes roughly 60 variance-ordered components, and even the full 549-dimensional span of the benchmark's activations covers only 0.79. Blend: under 5-fold CV, mixing the unit probe direction with the feature leaves the refusal-vs-wrong AUC flat at 0.944 for mixing weights up to 0.2 and degrades monotonically beyond, and a two-feature logistic stack over both projections also lands at 0.943; the feature's out-of-subspace mass carries no class information the probe has not already extracted. The raw projection onto 2191 alone reaches AUC 0.895, close to the probe's 0.944: the single dictionary direction carries nearly probe-grade class information in the probed representation.

D.4 Additional refusal-direction diagnostics

Score histograms (Section 6.1). Figure 11 shows the signed projections of the layer-28 hidden states onto the recovered refusal direction.

Refusal-direction lens scores (Section 6.1). The lift $(\mathbf{w}_{\text{raw}})_i = (V^\top w)_i/\sigma_i$ is the gradient of the probe score with respect to the raw state, the hidden-state direction along which the score increases fastest; raw units matter because the model's RMSNorm/LM head and the Section 6.2 intervention hook know nothing about the scaler. The lens scores are relative alignment scores, not log-probabilities: with tied embeddings, $\ell_v = \langle \mathbf{e}_v, \mathbf{g} \odot \hat{\mathbf{w}} \rangle$ (token row $\mathbf{e}_v$, RMSNorm gain $\mathbf{g}$, $\hat{\mathbf{w}}$ the

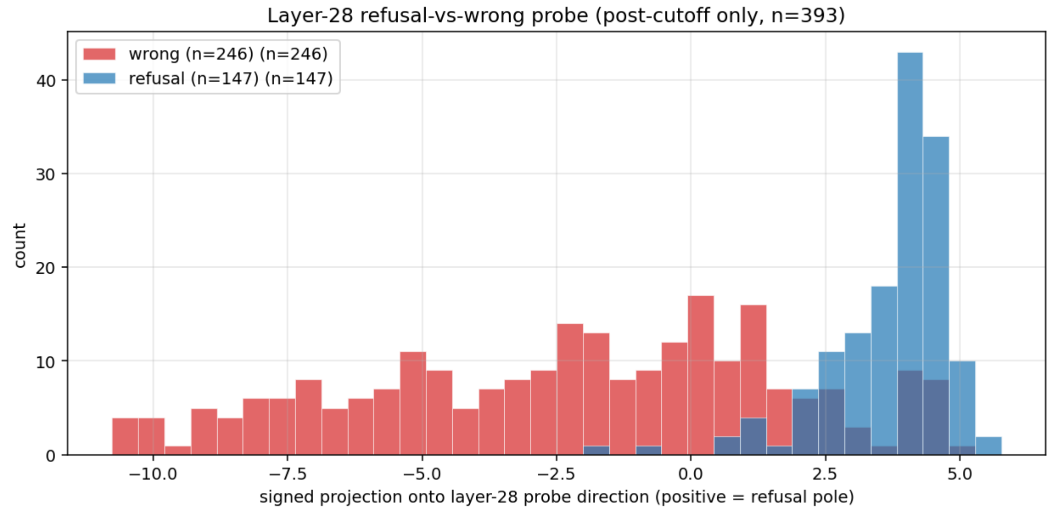


Figure 11: Score histograms: the signed projection of each item's layer-28 hidden state onto the recovered refusal direction, on the full refusal+wrong subset (top) and the within-post subset (bottom). Refusals concentrate at large positive projections; wrongs near zero or negative.

direction divided by its root-mean-square). Full subset ($n = 549$): `as` leads at $+19.22$, the Chinese refusal openers 作为一个 and 作为 (“as” / “as a”) sit at ranks 2–3 ($+17.67$, $+17.14$, with further variants at ranks 12–13), the rest of the top ten are case and spacing variants, and `there` at rank 11 scores $+14.16$. The within-post recovery's lens yields the same vocabulary in a sharper form: ” as” scores $+31.66$, “作为” $+29.63$, “\tas” $+27.58$, ” there” $+23.26$. The confabulation pole of the refusal direction is dominated by generic low-frequency tokens with no thematic structure, as expected: a confabulation can begin with any entity name, so there is no concentrated “confabulation vocabulary.” The layer-18 correctness direction under the same lens yields generic low-frequency fragments at the correct pole and at most a faint uncertainty flavor (不知 “don't know”, `unknown`, `forgotten`) at the incorrect pole.

Recovery-method check: difference in class means (Section 6.1). The raw difference of class means, $\bar{h}_{\text{refusal}} - \bar{h}_{\text{wrong}}$, at the same layer is *not* the probe direction: cosine 0.46 on the full subset (0.36 within post-cutoff, against a chance level of ≈ 0.02 in 2048 dimensions), with only one of its top-ten lens tokens (`as` itself, ranked first) overlapping the probe's. Its lens instead mixes the opener with a temporal-currency vocabulary (`current`, `currently`, `present`, 我没有 “I don't have”), which dominates outright within post-cutoff. The two recoveries weight different components: the mean difference absorbs topical and temporal-currency content, the discriminatively trained probe concentrates the register-and-opener component. A plain average of refusal states, as a control, points at nothing but generic sentence-starters (`As`, `The`, `There`, `In`).

Figure 2 shows the geometry behind the disagreement: the mean difference is by construction the centroid connector, with 96% of its norm in the top-two-PC plane where the wrong class's covariance is widest, while the probe direction keeps only 47% of its norm there, approximating instead the covariance-whitened difference $\Sigma^{-1}(\bar{h}_{\text{refusal}} - \bar{h}_{\text{wrong}})$ that separates the classes item by item.

The probe–2191 plane (Section 6.3.1). Figure 12 shows the plane spanned by the probe refusal direction and feature 2191's decoder vector.

From h-space to a-space: what a push looks like in features (Section 6.3). The three views of Section 6.3 are static, and the encoder $a(h)$ is heavily nonlinear (TopK), so how an h-space push maps into a-space is an empirical question. Figure 13 tracks four ensemble members' mean activations on the $n = 402$ wrong items as the post-block-27 residual is pushed along the Section 6.2 direction and along $\hat{W}_{\text{dec}}[2191]$. Three regularities emerge. First, 2191's activation grows smoothly and linearly under both pushes (from its baseline of 93 on wrong items to full coverage by $\alpha \approx +1000$ on the probe push and $+200$ on its own), with a probe-to-2191 slope ratio of 0.25, matching the two directions' cosine of 0.25: the TopK encoder is locally linear along these rays. Second, the two pushes differ in what else they recruit: the probe push feeds the whole register (17077 rises to a mid-dose peak, 4314 grows steadily), while the pure 2191 push crowds the hedge feature out entirely (4314

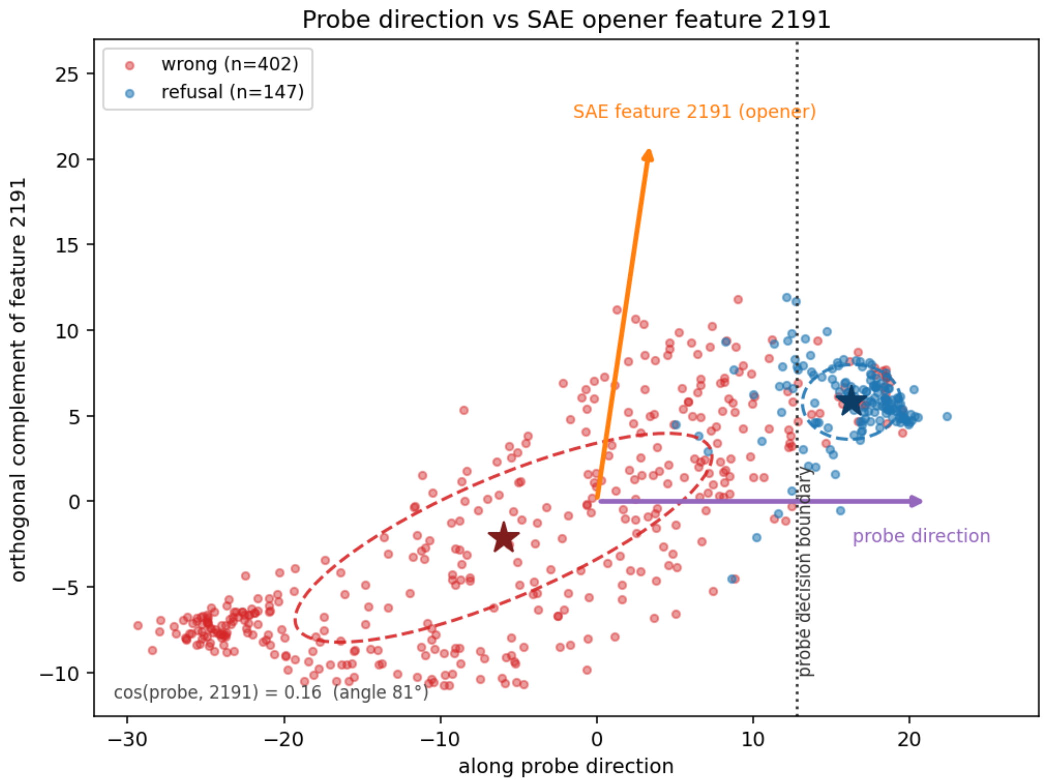


Figure 12: The plane spanned by the probe refusal direction and SAE feature 2191's decoder vector (refusal+wrong states in the probes' representation, $n = 549$; stars: class centroids; dashed: per-class 1σ covariance ellipses). The two arrows are 81° apart (cosine 0.16), yet the probe axis carries the class split and its decision boundary (dotted) is exact in this plane: a nearly probe-orthogonal output knob and a broad discriminative mixture share one causal channel.

falls to 0 by $+3000$). The probe direction moves wrong items *toward* refusal-like codes; the 2191 direction manufactures opener-dominated ones; behaviorally the two are indistinguishable (Appendix C), so the ensemble recruitment is not required for the flip. Third, the code as a whole is substantially rewritten at behavioral doses: active-set overlap with the unperturbed code falls to 0.36 (probe) and 0.26 (2191) at $\alpha = +1500$, and negative doses extinguish all four features.

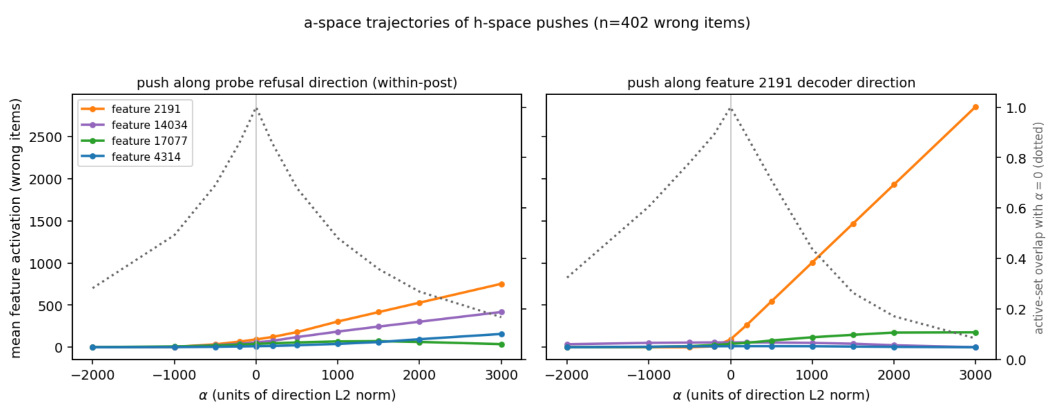


Figure 13: a-space trajectories of h-space pushes ($n = 402$ wrong items, pre-norm residuals). Solid lines: mean activation of four ensemble features as the state is pushed along the Section 6.2 refusal direction (left) or feature 2191's decoder direction (right); dotted: mean overlap (Jaccard) of the active feature set with the unperturbed code. The probe push recruits the register broadly; the 2191 push concentrates on the opener and crowds out the hedge.

D.5 Cross-model comparison table and per-layer probe curves

Table 16 is the Section 7.1 cross-model comparison summary. The correctness probe peaks in the same early-to-mid depth band on all three architectures, but the absolute peaks hide the confound

Section 5.6 exposed: a large fraction of what looks like model-internal correctness knowledge is the annotator-visible difficulty signal being read back through the hidden state, and the margins remain modest within the pre-cutoff stratum.

Table 16: Cross-model comparison summary.

Metric / Probe	Qwen3-1.7B	Gemma 2 2B	Llama 3.2 3B
Overall Accuracy	30.0%	31.9%	30.6%
Pre-cutoff Accuracy	47.3%	69.6%	80.7%
Post-cutoff Accuracy	19.5%	9.0%	0.2%
Refusal Rate on Post-cutoff Failures	37.4% (147/393)	55.2% (245/444)	97.5% (475/487)
Probe accuracies (peak % [peak layer] vs baseline)			
Correctness (all items)	**82.4%** [L18](base 70.0%, +12.4%)	**89.7%** [L14](base 68.1%, +21.6%)	**94.3%** [L13](base 69.4%, +24.9%)
Cutoff (all items)	**98.2%** [L13](base 62.2%, +36.0%)	**98.9%** [L14](base 62.2%, +36.6%)	**99.4%** [L25](base 62.2%, +37.1%)
Refusal-vs-Wrong (subset)	**89.4%** [L28](base 73.2%, +16.2%)	**83.9%** [L19](base 53.6%, +30.3%)	**95.8%** [L28](base 91.9%, +3.9%)
Correct within Pre-cutoff	**84.8%** [L18](base 52.7%, +32.1%)	**91.2%** [L17](base 69.6%, +21.6%)	**85.2%** [L13](base 80.7%, +4.4%)
Correct within Obscure	**80.5%** [L7](base 66.7%, +13.8%)	**88.9%** [L24](base 64.1%, +24.9%)	**81.1%** [L13](base 75.2%, +5.9%)
Prompt-feature baselines (strongest; margin vs probe above)			
Correctness vs +category	80.0% (+2.4)	88.0% (+1.7)	91.8% (+2.4)
Refusal-vs-Wrong vs TF-IDF	82.0% (+7.4)	67.6% (+16.3)	93.6% (+2.2)
Within-Pre Correct vs +category	81.8% (+3.0)	86.1% (+5.1)	79.4% (+5.8)

Margins are computed from unrounded values, so displayed operands can differ in the last digit (e.g. Llama correctness: $94.259 - 91.837 = +2.4$).

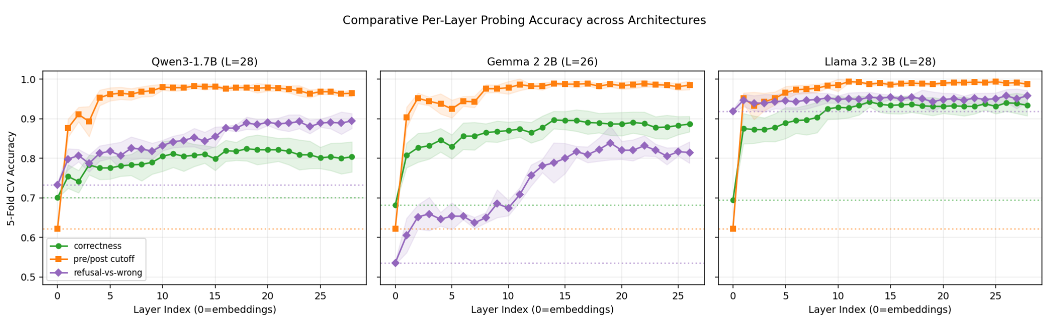


Figure 14: Comparative probing accuracy curves across Qwen3-1.7B, Gemma 2 2B, and Llama 3.2 3B (Section 7.1).

D.6 Refusal-vs-wrong probe under class imbalance (Section 5.5)

Bare accuracy (0.894) is misleading for the `refusal_vs_wrong` target because refusals are $147/549 = 26.8\%$ of the wrong+refusal subset: a constant predictor of `wrong` already scores 0.732. At the peak layer (28) the class-imbalance-aware metrics are: confusion matrix $[[\mathrm{TN}, \mathrm{FP}], [\mathrm{FN}, \mathrm{TP}]] = [[369, 33], [25, 122]]$, balanced accuracy 0.874, recall on refusals $122/147 = 0.830$, recall on wrong $369/402 = 0.918$, ROC AUC 0.944. The probe is not defaulting to "wrong": it flags 122 of 147 refusals at a false positive rate of $33/402 \approx 8.2\%$, and the AUC shows it ranks refusals well above confabulations in score order.

D.7 SAE feature-ranking views and the wrong-pole search (Section 6.3)

Figure 15: Four members of the refusal-register ensemble at the post-block-27 residual. Left column: the canonical refusal-opener (the Section 6.3.1 causal anchor) and the `Sorry/Oops` register. Right column: deferential officialese and the `moment` hedge. Hit rates and max-activating prompts are computed at the SAE's declared hook.

The three Section 6.3 views, for each of the $32{,}768$ features: **(A) direct encoding**, the features the encoder allocates to the direction placed at typical state magnitude (SAE.encode($\overline{\|h\|}\,\hat{w}_{\text{refusal}}$); the encoder is scale-sensitive); **(B) decoder alignment**, $W_{\text{dec}} \cdot w_{\text{refusal}}$ per feature, independent of the encoder nonlinearity; and **(C) empirical activation differential**, the standardized gap between mean activation on refusal items and wrong items, independent of the recovered direction. Convergent features are the most interpretable. Ensemble-member detail behind the Section 6.3 roster: the `moment` hedge 4314 (diff-z $+2.15$, the strongest discriminator) fires on 82% of refusals vs 12% of wrongs, and its max-activating prompts are post-cutoff prize questions the model refuses; officialese 17077 (`hereby`, `respectfully`, `duly`, `pursuant`) fires on 97% of refusals vs 40% of wrongs; the polite-modality feature 14361 reads `may`, `Please`, `Hopefully`; the first-person feature 8875 reads `I`, 我; 14034 is a register feature whose output vocabulary always loses the argmax race to "As"; and 2191's max-activating prompts are the recent-date items the model declines (Nov 2024 Gwen Stefani album, the *Wicked* adaptation, Taylor Swift album dates).

The strongest wrong-vs-correct feature in the Section 6.3 mirror search is a *topic* cue, not a commitment marker: an award-vocabulary detector (lens `winning`, 荣获, 获得) firing on 22% of wrong items vs. 1% of correct and 0% of refusals, whose max-activating prompts are the obscure Best Picture years of the model's weakest cell; it tops the disconfounded within-pre ranking too (diff-z $+1.34$). The runners-up are of the same species (a Nobel-Prize topic feature, an "In [date]" phrasing feature, a question-form feature), and against refusals the top wrong-side features are attempt and formatting features that fire on correct and wrong items alike. What the search rules out is a concentrated confabulation marker in this dictionary, the mirror image of 2191.

E. Bootstrap protocol and attribution details

Supporting protocol and accounting detail for Sections 8.1–8.3.

Table 17 is the Section 8.1 bootstrap table in full; Figure 16 is its forest plot.

Bootstrap protocol detail (Sections 4, 8.1). Per cell, $n_{\text{class}} = \min(|\text{pos}|, |\text{neg}|, 400)$; each subsample's h_{adds} is probe peak $-$ max(baselines) in %, the reported $\bar{h}_{\text{adds}}$ is the mean across K, and the 95% CI is the percentile interval $[h_{(0.025K)}, h_{(0.975K)}]$. $K = 30$ is set by compute (each replicate refits the full 29-layer $\times$ 5-fold pipeline), and percentile CIs at $K = 30$ are correspondingly coarse. The protocol is "resampling-of-balanced-subsamples" rather than a true item-level bootstrap; the latter would require $K \cdot L \cdot F \approx 4{,}350$ refits per cell, too slow on M1-class hardware (see the evaluation-methodology limitation in Section 10).

Table 17: Bootstrap 95% CIs on h_{adds} across the 14 (dataset, model, target) cells; forest plot in Appendix E (Figure 16).

dataset	model	target	n/class	$\bar{h}_{adds}$	95% CI	excl 0?
ConfabQA	Qwen3-1.7B	correct (all)	235	$+4.30$	$[+1.70, +7.45]$	**yes**
ConfabQA	Qwen3-1.7B	within_pre	140	$+1.25$	$[-1.07, +3.57]$	no
ConfabQA	Qwen3-1.7B	within_obscure	51	$+0.95$	$[-2.05, +6.90]$	no
ConfabQA	Gemma 2 2B	correct (all)	250	$+4.98$	$[+2.40, +7.20]$	**yes**
ConfabQA	Gemma 2 2B	within_pre	90	$+2.54$	$[-0.56, +5.00]$	no
ConfabQA	Gemma 2 2B	within_obscure	55	$+2.76$	$[-0.91, +6.36]$	no
ConfabQA	Llama 3.2 3B	correct (all)	240	$+0.94$	$[+0.21, +1.88]$	**yes**
ConfabQA	Llama 3.2 3B	within_pre	57	$+11.43$	$[+5.18, +19.45]$	**yes**
ConfabQA	Llama 3.2 3B	within_obscure	38	$+12.26$	$[+2.67, +23.67]$	**yes**
PopQA	Qwen3-1.7B	correct (full)	354	$+4.35$	$[+2.11, +6.50]$	**yes**
PopQA	**Qwen3-4B** (within-family scaling)	correct (full)	129	$+5.77$	$[+2.34, +11.61]$	**yes**
TriviaQA	Qwen3-1.7B	correct (full)	400	$+9.57$	$[+5.13, +13.75]$	**yes**
PopQA	**Llama 3.2 3B**	correct (full)	131	$+\mathbf{24.94}$	$[+\mathbf{20.57}, +\mathbf{29.03}]$	**yes**
TriviaQA	**Llama 3.2 3B**	correct (full)	356	$+\mathbf{21.25}$	$[+\mathbf{19.52}, +\mathbf{22.89}]$	**yes**

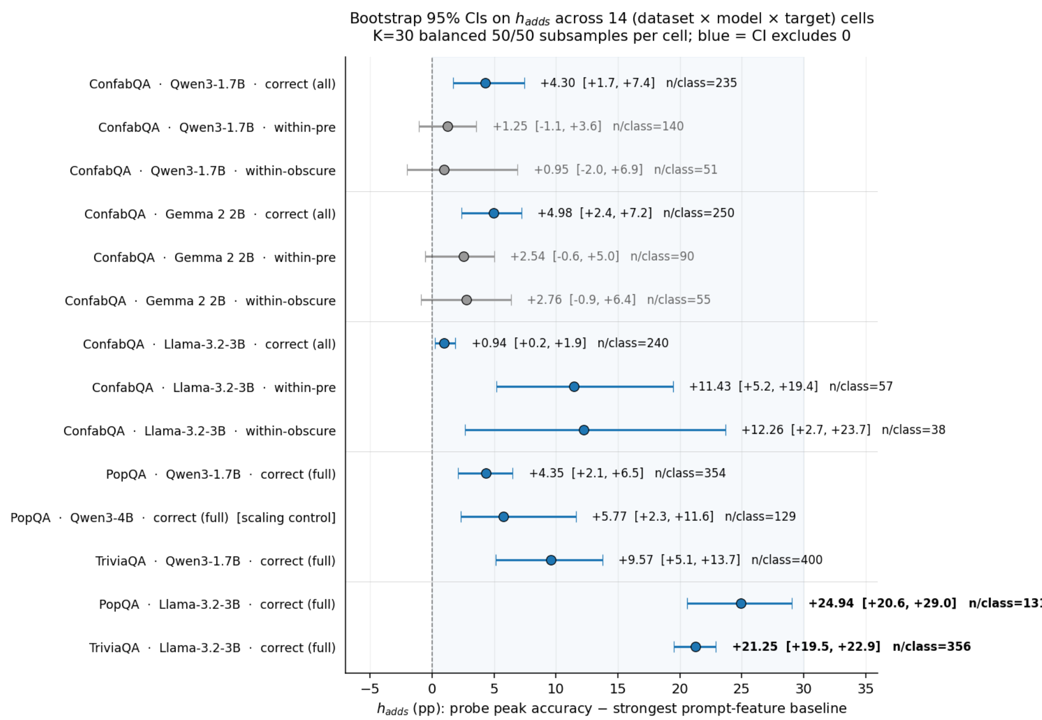


Figure 16: Forest plot of the Table 17 bootstrap 95% CIs on h_{adds} across 14 (`dataset, model, target`) cells. $K = 30$ balanced 50/50 subsamples per cell; blue = CI excludes 0. The bottom two rows are the headline positive finding: Llama 3.2 3B on PopQA / TriviaQA recovers $+21$–$+25\%$ over the strongest prompt-feature baseline, where Qwen3-1.7B and Qwen3-4B on the *same* data recover $+4$–$+10\%$. The Qwen3-4B "scaling control" row rules out parameter count as the dominant driver of the cross-model gap.

Permutation-null and item-level-bootstrap calibrations (Section 8.1). Two protocol checks calibrate the Table 17 CIs (artifact `small_cell_stats`; fixed layer, TF-IDF baseline, so absolute values are not comparable to the table). First, a label-permutation null of the peak-over-all-layers margin measures the layer-selection bias directly: null mean $+0.6\%$, 95th percentile $+1.3\%$ on both models, so margins in the $\approx 1\%$ range (e.g. Llama ConfabQA correct, $+0.94$) should not be read as evidence, while the all-items margins sit far above the null ($p < 0.01$). Second, an item-level bootstrap ($B = 400$) keeps the Llama all-items margin positive (CI excluding zero) but leaves *both* Llama within-stratum cells consistent with zero (within-pre $[-2.0, +8.5]$, within-obscure $[-3.3, +11.8]$):

the subsample protocol's positive CIs for those two cells understate item-sampling uncertainty and should be read as point estimates.

External-dataset sampling (Section 4). Both external datasets run through the `02_evaluate.py` generation pipeline. PopQA: 14k Wikidata triples templated into natural-language questions; sampled $n = 800$ stratified by `o_pop` (object popularity, 5 quintile bins of 160 items each), reshuffled at `seed` $\in \{0, 1, 2\}$. TriviaQA: `mandarjoshi/trivia_qa unfiltered.nocontext` validation split; sampled $n = 800$ uniformly, reshuffled at `seed` $\in \{0, 1, 2\}$. Generation settings match ConfabQA: greedy decoding, `enable_thinking=False`, last-prompt-token hidden-state capture at every layer.

Transfer-grid mechanics (Section 8.3). The fitted pipeline is transferred with its scaler and PCA held fixed at the source-dataset statistics; only evaluation touches the target dataset. The sweep uses a fixed four-layer grid per model (Qwen $\{14, 18, 22, 28\}$, Gemma $\{6, 13, 20, 26\}$, Llama $\{7, 14, 21, 28\}$); the Section 8.3 headline rows report the grid layer closest to each model's Table 16 peak (Qwen 18, Gemma 13, Llama 14), and the full grids are in the released matrices. A prompt-feature-only baseline (TF-IDF on question text) is fit on the same splits as a control: if the hidden-state probe transferred worse than the prompt-feature classifier, the hidden state would have contributed nothing portable beyond what the question text already encoded.

Transfer results in full (Section 8.3). Table 18 is the headline retention summary; the per-model readings follow.

Table 18: Cross-dataset transfer at each model's peak-within layer: margin retention without refit.

Model	Peak layer	Avg within margin	Avg transfer margin	Retention	Transfers > majority
Llama 3.2 3B	14	$+18.9\%$	$+15.2\%$	81%	$5/6$
Gemma 2 2B	13	$+13.1\%$	-7.0%	-54%	$4/6$
Qwen3-1.7B	18	$+8.7\%$	-1.2%	-14%	$3/6$

Llama: content-invariant. Five of six off-diagonal transfers beat the test-dataset majority, with an average winning margin of $+19.8\%$, comparable to the within-dataset $+18.9\%$. The single failure is ConfabQA $\to$ PopQA (majority 87.3%, transfer 79.8%). Two transfer cells (PopQA $\to$ ConfabQA at 94.0%, TriviaQA $\to$ ConfabQA at 93.8%) actually exceed the within-ConfabQA CV accuracy of 93.5%, suggesting less overfit when trained on more diverse data than ConfabQA alone provides.

Gemma: strong within, collapsed transfer. Gemma has the *largest* average within-dataset margin of the three at this layer but the worst transfer. Two cells collapse catastrophically, both with PopQA as target: ConfabQA $\to$ PopQA at 38.3% (vs majority 80.2%, a -42% gap) and TriviaQA $\to$ PopQA at 46.8% (-33.5%). Four cells do beat majority (transfers to ConfabQA and TriviaQA), average winning margin $+8.3\%$, well below Llama's $+19.8$; the net signed average is -7.0%, dominated by the PopQA collapses.

Qwen: weak within, weak transfer. Qwen3-1.7B's within-dataset margins are the smallest to begin with ($+8.7\%$ average); three of six transfers beat majority, all by small amounts ($+1.7$ to $+7.7\%$). Its signed transfer average (-1.2 pp) is closer to zero than Gemma's not because Qwen transfers better but because an almost flat probe stays almost flat under transfer (retention -14%).

Refusal census (Section 8.2) and Qwen3-4B raw accuracies (Section 8.1). Pool-level refusal rates behind the Section 8.2 anchor: Llama PopQA 62.9% vs. Qwen 1.7% ($39/2264$ refusals in the pooled three-seed Qwen sample); Llama TriviaQA 31.5% vs. Qwen 1.2% ($28/2242$). The four attribution cells are Qwen $\times$ {PopQA, TriviaQA} and Llama $\times$ {PopQA, TriviaQA}. Qwen3-4B's PopQA judge breakdown (14.8% correct / 1.6% refusal / 83.6% wrong) is essentially Qwen3-1.7B's, and the family's $\approx 2\%$ refusal rate versus Llama's 63% on the same items is the conspicuous family-level difference.

Refusal-channel attribution: full results (Section 8.2). Table 19 gives both tests in full.

(Italicized CI lower bound: 95% CI includes 0 because the refusal count is too small for the $K = 30$ percentile bootstrap to be well-resolved.)

Reading Test A. Comparing the drop-refusals rows to the unfiltered Table 17 rows: for Llama, $+24.94 \to +20.76$ on PopQA and $+21.25 \to +17.71$ on TriviaQA, i.e. the refusal channel accounts for $\approx 17\%$ of the headline and the bulk ($\approx 83\%$) is real correct-vs-wrong discrimination on attempted items. The refusal-channel hypothesis is *partially* right (about a sixth of the signal) but cannot account for the gap with Qwen, which would require nearly all of it. The Qwen rows are

Table 19: Refusal-channel attribution: Test A (drop refusals) and Test B (probe refusal directly).

Test	dataset	model	n/class	probe acc	strongest base-line	h_{adds}	95% CI
A: drop refusals	PopQA	Qwen3-1.7B	355	79.18%	75.87%	+3.31	$[+1.55, +6.48]$
A: drop refusals	PopQA	Llama 3.2 3B	102	82.37%	61.61%	**+20.76**	$[$**+14.67**$,$ **+27.94**$]$
A: drop refusals	TriviaQA	Qwen3-1.7B	400	69.40%	60.04%	+9.36	$[+6.50, +12.87]$
A: drop refusals	TriviaQA	Llama 3.2 3B	179	74.23%	56.52%	**+17.71**	$[$**+13.43**$,$ **+21.78**$]$
B: probe refusal	PopQA	Qwen3-1.7B	39	86.27%	76.04%	+10.24	$[0.00, +20.42]$
B: probe refusal	PopQA	Llama 3.2 3B	297	84.47%	66.03%	**+18.44**	$[$**+16.16**$,$ **+21.72**$]$
B: probe refusal	TriviaQA	Qwen3-1.7B	28	73.34%	57.37%	+15.97	$[-3.48, +32.27]$
B: probe refusal	TriviaQA	Llama 3.2 3B	252	90.23%	64.80%	**+25.43**	$[$**+22.81**$,$ **+29.76**$]$

essentially unchanged ($+4.35 \rightarrow +3.31$ PopQA, $+9.57 \rightarrow +9.36$ TriviaQA): Qwen refuses too rarely on these benchmarks for refusal-readout to be a substantial confound either way. One caveat on the ratio: its numerator is judge-labeled and its denominator substring-labeled (Appendix E). A label-sensitivity rerun (single pipeline, both regimes; artifact `small_cell_stats`) shows the contrast is not a label artifact: under judge labels Llama's TriviaQA margin grows ($+17.5 \rightarrow +23.8$ in that pipeline) and the Llama-vs-Qwen ordering is unchanged on both external datasets.

Reading Test B. Llama's refusal probe reaches 84.5% (PopQA) and 90.2% (TriviaQA) absolute accuracy on a balanced 50/50 task, beating the strongest prompt baseline by $+18.4$ and $+25.4\%$, CIs excluding 0 by wide margins: the hidden state encodes the abstention decision in a form the question text does not predict. Qwen's analogous margins ($+10.2\%$ PopQA, $+16.0\%$ TriviaQA) are positive but the refusal counts (39 and 28) leave the CIs straddling 0; this test needs either a larger Qwen evaluation or a benchmark on which Qwen3 refuses more (Qwen refuses 37.4% on ConfabQA's post-cutoff stratum).

Label provenance (Section 8.1). For the ConfabQA cells the `correct` field is the three-way judge's label (Section 5); for the external-dataset cells it is the generation-time substring match against gold and alternatives (the judge pass post-dates these bootstrap runs). Section 8.2's attribution tests use the judge labels instead, which is why their per-class counts differ from Table 17's (e.g. Llama PopQA: 131 substring-correct vs. 102 judge-correct).

Accuracy and baseline accounting for the Section 8.1 contrast. PopQA substring-correct rates are nearly identical (16.4% Llama vs. 15.6% Qwen); on TriviaQA Llama is more accurate (44.5% vs. 27.4%), but 50/50 balancing removes any base-rate advantage. Llama's strongest PopQA dropped-refusals prompt baseline is 61.6% vs. Qwen's 75.9%, so a lower baseline floor does not explain the gap either.

F. Extended discussion

The Section 9 readings in full.

The robust positive result. The refusal direction's logit lens recovers the literal refusal-opening vocabulary in each model (Qwen3: `as`, `As`, 作为; Gemma: `Regret`, `Formal`, `Alas`; Llama: `I`, `Given`, `As`), the one-shot intervention drives the first-token opener rate to 100% on Qwen3 and Gemma (Sections 6.2, 7.2), and the Section 8.2 refusal probe extends the finding to PopQA and TriviaQA at Llama scale; the claim "refusal is a clean, late-network, linearly-decodable, causally-load-bearing direction in these three small LMs" survives the bootstrap and the external benchmarks, and is no supervised artifact, being visible in unsupervised PCA (Figure 5b, Figure 6). The Qwen3-1.7B correctness direction fails all three corresponding tests: little margin over the difficulty oracle (Table 7), no token-space signature, no sign-directional causal effect (Section 6.2).

Why two atlases on Llama, one on Qwen. A plausible mechanism: Llama's heavier calibrated-abstention training requires a distinguishable internal "I should hedge" state (the Test-B gap) and a richer correctness representation as its *prerequisite* (what Test A recovers); Qwen3-1.7B, which prefers to attempt-and-confabulate, never had pressure to develop either. The Qwen3-4B scaling control supports this: same family, same near-zero PopQA refusal rate, h_{adds} essentially at Qwen3-1.7B's level and far below Llama's. The mechanism is family / post-training recipe, not scale; a fully clean ablation would still need a recipe swap holding family fixed, which is future work.

Implication for the probing literature. Probing papers reporting "the hidden state predicts correctness at ~ 80–90%" should be re-evaluated against a prompt-feature baseline (removing the cutoff variable is necessary but not sufficient), and the result is sharply model-dependent: a single-model study could report either "substantial factual self-knowledge" (Llama) or "nothing over prompt features" (Qwen ConfabQA within-obscure) on identical data and code, consistent with Orgad et al.'s (2024) finding that truthfulness encoding is multifaceted rather than universal. Headline small-LM claims should be checked on at least two architectures with different post-training recipes, with bootstrap CIs on balanced subsamples rather than single-seed point estimates. Finally, the refusal direction is a structure of the continuous representation space that becomes text one token later; a natural extension is to measure calibration directly in that space (signed distances to the refusal and correctness hyperplanes as continuous confidence scores).